

Linear Models in Animal Breeding

A Worked Approach

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Welcome

! Under Construction

This book is being written, and most of it is not finished yet. Chapter 1 is drafted; every other chapter and appendix is a placeholder showing only its title and learning objectives, so you can see the shape of the whole book and follow along as it fills in. Anything you read here may change. Do not cite it yet.

Welcome to **Linear Models in Animal Breeding: A Worked Approach**.

About This Book

This book provides a hands-on, example-driven introduction to mixed model equations for graduate students in animal breeding and genetics. Each chapter starts with small datasets (5-10 animals), demonstrates hand calculations step-by-step, and then scales up to realistic applications using R.

Learning Approach

Our pedagogical philosophy emphasizes:

- **Start Small:** every model begins with a toy dataset of five to ten animals that you can solve by hand
- **Numbers Before Symbols:** the operation is performed on real numbers first, and the general formula arrives afterwards as a name for something you have already done — never before it
- **Show Every Step:** every symbol is introduced as a name for something already on the page, and the dimensions of every matrix are stated
- **Hand First, Then R:** solve it with a calculator, then reproduce the identical answer in R, then see the one-line package call that does the same thing
- **Code You Can Read:** the R is shown, not folded away, and is written to be retyped
- **Check Yourself:** short predict-then-check questions, placed where you could have guessed the answer but have not been told it yet

What You'll Learn

By working through this book, you will:

- Understand the theory and practice of mixed model equations (MME)
- Master pedigree-based relationship matrices and their inverses
- Implement animal models for single and multiple traits
- Prepare data properly: contemporary groups, connectedness, and pedigree quality
- Model longitudinal data with random regressions and reaction norms
- Handle categorical traits with threshold models and survival analysis
- Fit non-additive, multibreed, and crossbred models

- Estimate variance components using ANOVA, REML, AI-REML, and Gibbs sampling
- Validate a genetic evaluation for accuracy, bias, and dispersion

Prerequisites

Students using this book should have:

- Elementary statistics (linear regression, variance components, distributions)
- R programming experience (helpful but not strictly required - all code is explained)

No prior linear algebra is assumed. Chapter 1 develops every matrix operation the book uses, and tells you which sections each later chapter depends on.

Scope

This book covers the **pedigree-based** mixed model. Genomic selection - the genomic relationship matrix, GBLUP, single-step, and the Bayesian alphabet - is a large enough subject to need its own volume, and is deferred to a companion book rather than compressed into a chapter here. Large-scale computation is likewise deferred, apart from Chapter 24, which exists so that readers understand why production evaluations are not solved the way the examples in this book are.

The chapters

Every chapter is listed below, and the sidebar on the left carries the same list on every page. **Chapter 6 is the centre of the book** — everything in Part I builds toward it, and everything after it is a variation on it.

Chapters marked are drafted. The rest are placeholders carrying their learning objectives only.

Part I — Background

	Chapter	What it does
1	Matrix Algebra for Animal Breeders	Only the linear algebra the book actually uses, with a map of what is needed where
2	Linear Models	Fixed effects only. Build $\mathbf{X}$, solve the normal equations, confront non-full-rank
3	Mixed Models Without Relationships	$\mathbf{Z}$ and random effects, with no genetics attached yet
4	Pedigrees and Relationship Matrices	$\mathbf{A}$ and $\mathbf{A}^{-1}$ — the object that makes the animal model work
5	Data Preparation and Connectedness	What goes wrong before a model is ever fitted. Where real analyses fail

Part II — Core Animal Models

	Chapter	What it does
6	The Animal Model	The central chapter. Solve the full MME with $\mathbf{A}^{-1}$, then interpret what came out

	Chapter	What it does
7	Equivalent and Reduced Models	Sire, sire-MGS, and reduced animal models as data reduction
8	Random Environmental Effects	Repeatability and common environment: a second random effect that is not genetic
9	Genetic Groups	Unknown parent groups, and an honest statement of what they get wrong
10	Maternal Effects Models	Two correlated genetic effects on one phenotype

Part III — Advanced Model Structures

	Chapter	What it does
11	Multivariate Models	Several traits at once; the Kronecker product earns its keep
12	Longitudinal Data and Random Regression	Traits measured along a trajectory
13	Social Interaction Models	An animal's phenotype depends on its group-mates' genes

Part IV — Categorical and Time-to-Event Traits

	Chapter	What it does
14	Threshold Models	Binary and ordered categorical traits — same model, more thresholds
15	Survival Analysis	Time-to-event data and censoring

Part V — Non-additive Genetic Effects

	Chapter	What it does
16	Dominance Models	Genotypic value, not just breeding value
17	Epistasis Models	Why, in practice, nearly everything comes out additive

Part VI — Multibreed and Crossbred Evaluation

	Chapter	What it does
18	Multibreed Models	More than one breed in a single evaluation
19	Crossbred Evaluation and CCPS	Selecting purebreds for crossbred performance

Part VII — Variance Component Estimation

	Chapter	What it does
20	ANOVA and Henderson's Methods	Where variance components came from, and why they had to be replaced
21	REML	The workhorse
22	Bayesian Estimation via Gibbs Sampling	The third route to the same components, and the one that gives full uncertainty

Part VIII — Validation and Computation

	Chapter	What it does
23	Validating Genetic Evaluations	How do you know the evaluation works?
24	Solving the MME at Scale	What production software actually does. Deliberately thin

Appendices

A — Notation Reference · B — BLUP Derivation · C — Inbreeding Algorithms · D — Legendre Polynomials · E — Deregression · F — R Code Reference · G — Datasets · H — Solutions to Exercises · I — BLUPF90 Guide

Appendix A is the one to bookmark. It is the authoritative notation reference and carries an index of every Key Equation in the book, one line each — the single page to revise from.

How to Navigate

Chapters 1-6 form the essential foundation and should be read in order. After that, you can follow different paths depending on your interests:

- **Linear path** (comprehensive): read chapters sequentially
- **Longitudinal models**: Ch 1-6, 8, 11-12, 21
- **Categorical and fitness traits**: Ch 1-6, 14-15, 22
- **Crossbreeding**: Ch 1-6, 9, 11, 16, 18-19
- **Variance components**: Ch 1-6, 20-22, 23

Each chapter lists its prerequisites at the start.

How to read a chapter

Every chapter is built the same way, so once you have read one you know where everything is.

The order is always the same. A chapter opens on a small dataset, works it in literal numbers, and only then writes the general form. A formula never arrives before the worked example that earns it. If you want the rule without the walk-through, skip to the boxed equation — but the numbers above it are where the understanding actually comes from.

Coloured boxes mean different things. The colour tells you what to do with the box:

Box	What it is
Key Equation	The one to memorise. One to three per chapter, and nothing else is boxed
Derivation	Optional, collapsed. Why the result holds in general. Skip it on a first read
Check Yourself	A five-second self-test. The question is visible, the answer is hidden — try it before opening
Common Mistake	The error students actually make, and how to catch it
In R	An R-specific trap or idiom
Notation Watch	Where the literature disagrees with itself
In Practice	What a real national evaluation does
Beyond This Book	A scope boundary, and where the topic is covered instead

Displayed equations come in three kinds. A boxed, numbered equation is one you should be able to write from memory. Ordinary display maths in the flow is something to follow with the book open. Anything inside a collapsed grey Derivation box is optional.

The R is meant to be typed. Code is shown by default, not folded away, and every chunk prints the same number the hand calculation produced so you can check the two against each other. Open an R session alongside the book and work down the page. The “Code” menu at the top right will download the source of any chapter.

Some sections have short animations. Where a calculation is a *sequence* — a row walking against a column, a matrix filling in cell by cell — there is a short silent clip showing the steps. They are never the only place something is explained: everything in a clip is also in the text, so nothing is lost if you skip them or are reading in print.

Datasets referenced in examples are available in the `data/` directory of the book repository.

Getting set up in R

The book is written to be read with an R session open beside it. Everything you need is free, and the whole setup takes about ten minutes.

1. Install R

R itself comes from **CRAN**, the Comprehensive R Archive Network:

- <https://cran.r-project.org/> — choose your platform (Windows, macOS, or Linux) and take the latest release

Any version from **4.3 onward** will run everything in this book. To check what you have, open R and run:

```
1 R.version.string
```

2. Install an editor

R on its own is a bare console. You want an editor that keeps your script, your console, your plots and your data in one window:

- **RStudio Desktop** — free, the standard choice, and what most course material assumes
- **Positron** — Posit’s newer editor, if you also work in Python
- **VS Code** with the R extension, if you already live there

Install R *first*, then the editor: the editor looks for an existing R installation when it starts.

3. Install the packages

Chapter 1 needs nothing at all. `Matrix`, `MASS` and `survival` are *recommended* packages — they ship with every standard R installation, so the whole of the matrix algebra chapter runs on a fresh install with no downloads.

Later chapters use a handful of specialist packages. Install them when you reach them, or all at once now:

```
1 install.packages(c(
2   "lme4",           # general mixed models (Ch 3)
3   "pedigreemm",     # pedigree-based mixed models (Ch 4, Ch 5)
4   "nadiv",          # relationship matrices and their inverses (Ch 4, Ch 16)
5   "sommer",         # REML for animal models, multi-trait (Ch 10, Ch 11, Ch 21)
6   "MCMCglmm",       # Bayesian mixed models via MCMC (Ch 14, Ch 22)
7   "coda",           # MCMC convergence diagnostics (Ch 22)
8   "coxme",          # survival models with random effects (Ch 15)
9   "orthopolynom",   # Legendre polynomials (Ch 12)
10  "tidyverse"       # data preparation and validation (Ch 5, Ch 23)
11 ))
```

Two of these — `sommer` and `MCMCglmm` — compile from source on some systems and can take several minutes each. That is normal; let them finish.

To check that a package installed correctly, load it:

```
1 library(sommer)
```

If that returns a prompt with no error, you are set.

4. Running the book's code

Every code chunk in this book is **shown, not hidden**, because the code is part of what is being taught rather than an appendix to it. Three ways to get it:

- **Copy one chunk.** Hover over any code block and a copy button appears in its top-right corner.
- **Take a whole chapter.** The `</> Code` menu at the top right of each page will show or hide every chunk at once, or download that chapter's source.
- **Clone the whole book.** If you want the datasets and the generating scripts as well:

```
1 git clone https://github.com/austin-putz/linear-models-in-animal-breeding.git
```

Then, from the project directory, `Rscript setup.R` installs the full package list in one go.

Work down the page rather than skipping to the end of a section: each chunk prints the same number the hand calculation above it produced, and comparing the two is the point of having the code there at all.

Citation

If you use this book in your research or teaching, please cite as:

Putz, Austin (2025). *Linear Models in Animal Breeding: A Worked Approach*. <https://austin-putz.github.io/linear-models-in-animal-breeding/>

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appropriate credit and distribute your contributions under the same license.

Feedback and Contributions

Found a typo, error, or have a suggestion? Please open an issue on GitHub or submit a pull request. This book benefits from community input and we welcome contributions.

Acknowledgments

Special thanks to the students who provided feedback on early drafts, and to the open-source community for the tools that made this book possible — Quarto, R, the **Matrix** package, and Manim for the animations.

Ready to begin? Start with Chapter 1: Matrix Algebra for Animal Breeders.

Part I

Part I: Background

Chapter 1

Matrix Algebra for Animal Breeders

Learning Objectives

By the end of this chapter, you will be able to:

1. **Perform** transpose, addition, multiplication, and inversion **by hand** on matrices up to 4×4 , and **verify** each result in R
2. **Diagnose** a singular matrix, **explain** why breeding models routinely produce one, and **use** a generalized inverse
3. **Apply** the Kronecker and Hadamard products and **name** the chapter where each is used
4. **Decompose** a covariance matrix by Cholesky and by eigenvalues, and **state** what each decomposition is for
5. **Derive** the variance of a linear combination, $\text{Var}(\mathbf{A}\mathbf{y}) = \mathbf{A} \text{Var}(\mathbf{y}) \mathbf{A}'$, and **use** it to assemble $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}' + \mathbf{R}$

No prerequisites — this is the entry point.

1.1 Why linear algebra?

Figure 1.1 is this chapter on one page. Nothing in it needs to make sense yet — come back to it when you have finished, and use it as a map in the meantime.

Two entries in the bottom row — the **genomic relationship matrix** and **single-step GBLUP** — are outside this book's scope and belong to the companion volume on genomics. They are in the figure so you can see where the pedigree-based methods you are about to learn eventually lead, not because they are taught here.

1.1.1 What it is

Linear algebra is the mathematics of many quantities at once. Ordinary algebra handles one unknown at a time: solve $4x = 12$ and you have $x = 3$. Animal breeding never asks a question that small. A genetic evaluation asks for the breeding values of every animal in a population *simultaneously*, because no animal's value can be worked out without its relatives' — and those depend back on it. That is not one equation. It is a hundred thousand equations that must all be true at the same time.

Linear algebra is the language for writing down such a system and solving it. It gives you two things ordinary algebra cannot:

1. **A way of writing that does not grow with the problem.** The same four symbols describe four animals or four million. You will see this happen in §1.7.

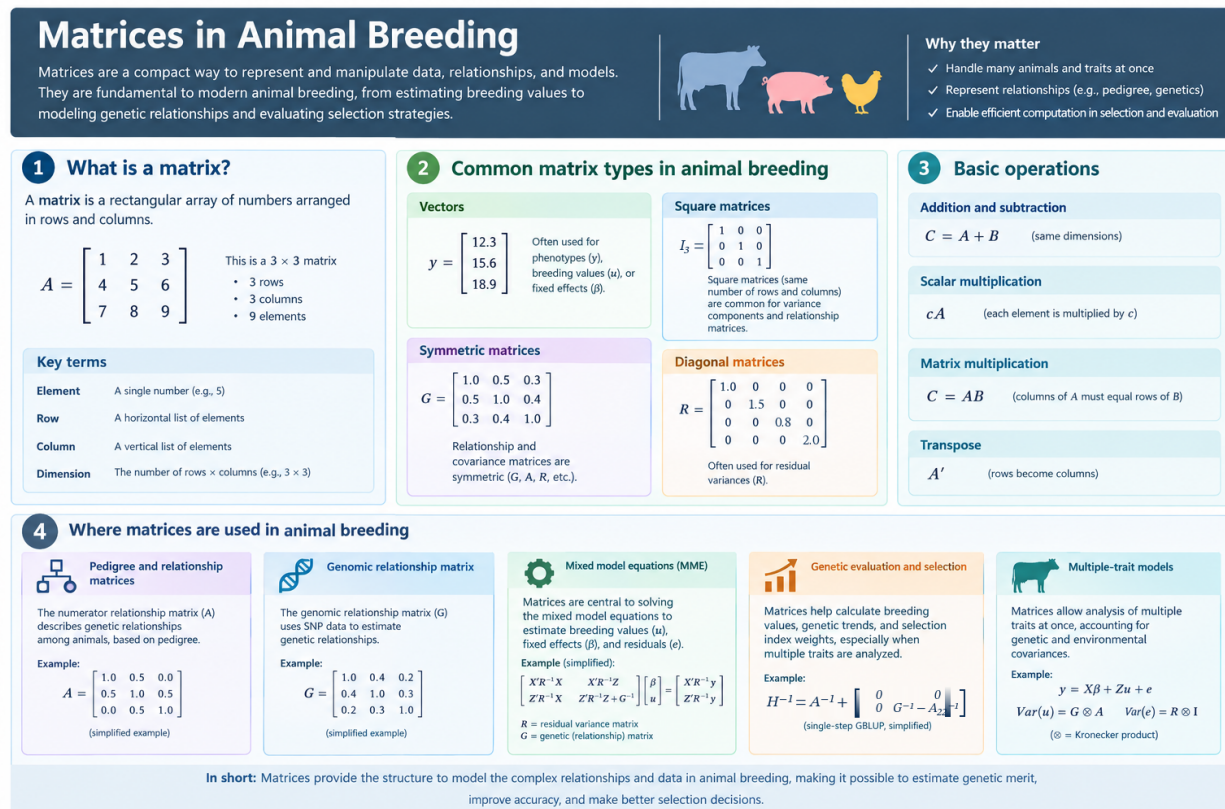


Figure 1.1: Matrices in animal breeding. Panel 1 is §1.2, panel 2 is §1.6, panel 3 is §1.3 to §1.5, and panel 4 is the applications map in §1.1.2 below.

2. **Operations on whole collections of numbers at once** — an arithmetic where the objects being added and multiplied are entire tables rather than single values.

The word *linear* is a real restriction and worth knowing about. It means every unknown appears only to the first power, multiplied by a constant and added to the others — no squares, no products of two unknowns, no unknowns inside a logarithm. Most of quantitative genetics is built to stay inside that restriction, because linear systems are the ones we can actually solve at scale. The threshold models of Chapter 14 are what it looks like when a trait refuses to fit, and what has to be done about it.

1.1.2 Where it appears in animal breeding

Almost every quantitative method a breeder uses is linear algebra with a different name on it. The table below is the map: the left column is where you meet the problem, the middle column is what the algebra actually has to do, and the right column is the section of this chapter that teaches it.

Where it shows up	What the linear algebra does	Taught in
The animal model (Ch 6)	Build the mixed model equations (Henderson 1975), solve them for every effect at once, and read accuracy off the inverted coefficient matrix	§1.5, §1.9, §1.15
Relationship matrices (Ch 4)	Build A from a pedigree and invert it — the object that makes the animal model practical (Henderson 1976)	§1.9, §1.12
Selection index (Ch 6)	Find the weights that combine an animal's own record with its relatives' by solving a linear system, $\mathbf{b} = \mathbf{P}^{-1}\mathbf{Ga}$ (Hazel 1943)	§1.9
Fixed effects and contemporary groups (Ch 2, Ch 5)	Decide which effects can be estimated at all, and which herds can be compared with which	§1.8, §1.10
Multiple traits (Ch 11)	Expand a small trait covariance matrix over every animal in the pedigree	§1.11
Random regression (Ch 12)	Fit a curve per animal from a polynomial basis, and reduce twenty traits to the few dimensions carrying the variation	§1.13
Variance components (Ch 20–22)	Build a likelihood out of quadratic forms, traces, and log-determinants, then differentiate it (Patterson and Thompson 1971)	§1.12, §1.14
Simulation (Ch 11, Ch 22)	Turn independent random draws into correlated breeding values with the right covariance structure	§1.12, §1.15
Solving at scale (Ch 24)	Get an answer when the coefficient matrix has a million rows and cannot be inverted directly	§1.9
Optimal contribution selection	Maximise genetic merit subject to a ceiling on average relationship — a quadratic form $\mathbf{x}'\mathbf{Ax}$ in the contributions	§1.14 · <i>theory in the companion volume on breeding programme design</i>

Where it shows up	What the linear algebra does	Taught in
Genomic prediction	Replace $\mathbf{A}$ with a relationship matrix built from markers. The surrounding algebra does not change	<i>companion volume on genomics</i>

Two things are worth noticing about that list before you start.

The same few operations keep reappearing. Multiply, transpose, invert, decompose. Optimal contribution selection and REML look nothing like each other on the page, yet both rest on a quadratic form — the object of §1.14 — and both need the same relationship matrix. Learning an operation once buys you every application of it, which is why this chapter is organised by operation rather than by application.

The last two rows are deliberately out of scope here. Optimal contribution selection belongs to breeding programme design and genomic prediction to genomics; each has a companion volume, and neither is taught in this book. They are in the table because the algebra underneath them is the algebra in this chapter — a student who works through Chapter 1 is equipped for both. ### Why not just run the software?

Every operation in this chapter is one line of R, and nobody computes a national evaluation by hand. So why learn to?

Because much of what goes wrong in a real evaluation is visible in the algebra before it is visible in the results — and sometimes never visible in the results at all. An effect that cannot be estimated still returns a number (§1.10). A genetic correlation of exactly 1 stops REML converging, and the error names a matrix rather than the pair of traits that caused it (§1.13). Two herds with no animals in common produce breeding values that look comparable and are not (Chapter 5). In each case the software is behaving correctly; the person reading the output is the one who has to recognise what happened.

The narrower reason is that Chapter 6 is unreadable without this material. Henderson’s equations are written in matrices, solved with a matrix inverse, and their accuracies read off a matrix diagonal. A student who cannot form $\mathbf{X}'\mathbf{X}$ cannot see what those equations are doing, and is left running software and trusting the output — which is exactly the position the paragraph above says you should not be in.

It is worth knowing how far this transfers, too. Regression, ANOVA, and the general linear model are the same computation as §1.9 with different words attached: `lm()` in R forms $\mathbf{X}'\mathbf{X}$ and solves it, using the drop-a-level generalized inverse of §1.10.

1.1.3 How to use this chapter

This is a reference. Read it once to see what is here, then come back to whichever section you need — §1.16 maps every section to the chapters that consume it. Every section opens with real numbers and works them before stating any rule, so landing in §1.14 without having read §1.4 still works.

Three small examples carry the whole chapter, and they are reused on purpose: the algebra changes while the animals stay the same, which is the only way to see that it is the algebra doing the work.

1.2 Matrices, vectors, and scalars

Before any operation, the objects themselves. Everything in this chapter is one of three things, and a great deal of confusion later comes from losing track of which.

1.2.1 The objects

Four lambs were measured for two traits — yearling weight in kilograms and loin muscle depth in millimetres:

Lamb	Yearling weight (kg)	Loin depth (mm)
1	48	25
2	52	28
3	41	22
4	45	25

Strip away the headings and what is left is eight numbers in a rectangle, four rows deep and two columns wide:

$$\mathbf{Y} = \begin{bmatrix} 48 & 25 \\ 52 & 28 \\ 41 & 22 \\ 45 & 25 \end{bmatrix}$$

That is a **matrix**, and it is the only object this chapter is really about. Nothing has been modelled, assumed, or estimated. The rectangle is just the data, arranged so that position carries meaning: row 3 is lamb 3, column 2 is loin depth, and the number where they meet — 22 — is lamb 3’s loin depth.

```

1 # The measurement table above, as a matrix. byrow = TRUE fills it the way the
2 # table reads: across row 1, then across row 2.
3 Y <- matrix(c(48, 25,
4               52, 28,
5               41, 22,
6               45, 25), nrow = 4, byrow = TRUE)
7 Y
8 #>      [,1] [,2]
9 #> [1,]  48  25
10 #> [2,]  52  28
11 #> [3,]  41  22
12 #> [4,]  45  25
13
14 dim(Y)      # rows first, always: 4 by 2
15 #> [1] 4 2
16 Y[3, 2]     # row 3, column 2 -- lamb 3's loin depth, 22
17 #> [1] 22
18
19 # One column on its own is a vector. We will use the weights repeatedly, so
20 # give them a name now.
21 y <- Y[, 1]
22 y
23 #> [1] 48 52 41 45

```

i Definition — Matrix, vector, and scalar

A **matrix** is a rectangular array of numbers in which position carries meaning. A **vector** is a matrix with a single column. A **scalar** is a single number.

Its **dimensions** are quoted rows first: $\mathbf{Y}$ above is 4×2 , read “four by two”. The number in row i and column j is called an **element** and written y_{ij} , again row first, so $y_{32} = 22$.

Why it matters. These three words are the whole vocabulary of the chapter, and the book uses typography to keep them apart: **bold uppercase** for a matrix ($\mathbf{Y}$, $\mathbf{X}$, $\mathbf{A}$), **bold lowercase** for a vector ($\mathbf{y}$, $\mathbf{b}$), plain italic for a scalar (σ_e^2 , α). When you meet $\mathbf{Y}$ and $\mathbf{y}$ in the same equation in Chapter 11, they are different objects and the case is the only thing telling you so.

Take one column of $\mathbf{Y}$ on its own — the four weights — and you have a vector, 4×1 :

$$\begin{bmatrix} 48 \\ 52 \\ 41 \\ 45 \end{bmatrix}$$

This is the shape almost everything in the book arrives in: one row per record, in a fixed order that never changes. That fixed order is what lets a second table — which flock each lamb is in, which sire it is by — line up against it row by row without anyone having to say so.

1.2.2 The one idea underneath all of it

Suppose you wanted a single index combining both traits, counting a kilogram of weight as worth 1 and a millimetre of depth as worth 2. For lamb 1 that is

$$1(48) + 2(25) = 98$$

Multiply each quantity by a weighting, add up the results. That operation — and *only* that operation — is what linear algebra automates.

i Definition — Linear combination

A **linear combination** of a set of quantities is what you get by multiplying each one by a constant and adding the results: $c_1x_1 + c_2x_2 + \dots + c_nx_n$. The constants are called **coefficients** or **weights**.

Why it matters. This is the single idea the whole chapter is built from, and recognising it saves you learning the same thing five times. Matrix multiplication (§1.5) is linear combinations performed wholesale. A fitted value $\mathbf{Xb}$ is a linear combination of effects. A selection index is a linear combination of trait records. A breeding value is a linear combination of the information on an animal and its relatives. Every one of them is “multiply by weights, then add” — what changes between them is only *where the weights come from*.

The same weights applied to every lamb at once, still one multiplication and one addition per term:

```
1 1 * Y[, 1] + 2 * Y[, 2]      # 98 for lamb 1, and the other three
2 #> [1] 98 108 85 95
```

Section 1.5 replaces that line with a single matrix product. Nothing about the arithmetic changes — only how much of it you have to write down.

1.2.3 Notation

These are conventions, not results. There is nothing to discover — they are names, and you should be handed them rather than led to them.

The book writes **bold uppercase** for a matrix, **bold lowercase** for a vector, and plain italic for a scalar, as set out above. Two more pieces of vocabulary before we start operating on them.

A matrix is **square** when it has as many rows as columns. Only square matrices have determinants, inverses, and eigenvalues, so a good deal of this chapter applies to them alone. Our $\mathbf{Y}$ is 4×2 and therefore not square — but $\mathbf{Y}'\mathbf{Y}$ would be, and it is no accident that the products at the centre of every model in this book come out square however many records you have.

The **main diagonal** of a square matrix runs from its top-left corner to its bottom-right: the elements $a_{11}, a_{22}, a_{33}, \dots$, the ones whose row and column index are the same. Almost everything interesting about a square matrix is either on that diagonal or defined relative to it.

1.3 Transpose

The simplest operation there is. It needs nothing but the matrix itself, which is why it comes first.

Write the measurement matrix $\mathbf{Y}$ from §1.2, and then write it again with its rows and columns interchanged:

$$\mathbf{Y} = \begin{bmatrix} 48 & 25 \\ 52 & 28 \\ 41 & 22 \\ 45 & 25 \end{bmatrix} \quad \mathbf{Y}' = \begin{bmatrix} 48 & 52 & 41 & 45 \\ 25 & 28 & 22 & 25 \end{bmatrix}$$

$\mathbf{Y}$ is 4×2 and $\mathbf{Y}'$ is 2×4 . Row 1 of $\mathbf{Y}$ — lamb 1's two measurements — has become column 1 of $\mathbf{Y}'$. The entry that sat in row 3, column 2 now sits in row 2, column 3. That is the whole operation, and written out it is simply

$$y'_{ij} = y_{ji}$$

The transpose is written $\mathbf{Y}'$ in this book and read “Y prime”; some texts write $\mathbf{Y}^T$ for the same thing. Transposing twice puts everything back, so $(\mathbf{Y}')' = \mathbf{Y}$.

```

1  t(Y)                                # 4 x 2 becomes 2 x 4
2  #>      [,1] [,2] [,3] [,4]
3  #> [1,]   48   52   41   45
4  #> [2,]   25   28   22   25
5  all.equal(t(t(Y)), Y)  # transposing twice returns the original
6  #> [1] TRUE

```

A square matrix that is unchanged by transposing is called **symmetric**: its element in row i , column j equals the one in row j , column i , so it is a mirror image of itself across the diagonal running from top-left to bottom-right.

$$\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

That matters more than it looks. **Every relationship matrix and every covariance matrix in this book is symmetric**, because the relationship between animal 1 and animal 2 is the same number as the relationship between animal 2 and animal 1. So is $\mathbf{X}'\mathbf{X}$ (§1.7), and so is the whole left-hand side of the mixed model equations in Chapter 6. It is a fact used constantly and almost never announced, and it is worth carrying: whenever you see $\mathbf{A}'$ written for a covariance matrix, it is the same matrix as $\mathbf{A}$.

1.4 Addition, subtraction, and scalar multiplication

Like the transpose, these three leave every number where it is. Nothing travels; position is preserved. That is exactly what makes them easy, and it is the property matrix multiplication gives up in §1.5.

1.4.1 Addition and subtraction

Matrix addition is element by element, position preserved. Here is the case worth caring about.

For sheep yearling weight (kg) and loin muscle depth (mm), suppose the additive genetic covariance matrix and the residual covariance matrix are

$$\mathbf{G} = \begin{bmatrix} 36 & 12 \\ 12 & 9 \end{bmatrix} \quad \mathbf{R} = \begin{bmatrix} 64 & 4 \\ 4 & 16 \end{bmatrix}$$

Call these **Example B**. They are the two-trait covariance structure the rest of the chapter returns to.

Adding them position by position — $36 + 64$, $12 + 4$, $12 + 4$, $9 + 16$ — gives the phenotypic covariance matrix:

$$\mathbf{P} = \mathbf{G} + \mathbf{R} = \begin{bmatrix} 100 & 16 \\ 16 & 25 \end{bmatrix}$$

Everything a breeder quotes comes off these three matrices by division. The phenotypic standard deviations are $\sqrt{100} = 10$ kg and $\sqrt{25} = 5$ mm. The heritabilities are the diagonal of $\mathbf{G}$ over the diagonal of $\mathbf{P}$: $36/100 = 0.36$ for weight and $9/25 = 0.36$ for depth. The genetic correlation is the off-diagonal of $\mathbf{G}$ over the square root of the product of its diagonals, $12/\sqrt{36 \times 9} = 12/18 = 0.67$; the residual correlation is $4/\sqrt{64 \times 16} = 0.125$; the phenotypic correlation is $16/(10 \times 5) = 0.32$.

```

1 G <- matrix(c(36, 12,
2               12,  9), nrow = 2)  # additive genetic covariances
3 R <- matrix(c(64,  4,
4               4, 16), nrow = 2)  # residual covariances
5 P <- G + R                        # phenotypic covariances, element by element
6 P
7 #>      [,1] [,2]
8 #> [1,] 100  16
9 #> [2,]  16  25
10
11 # Heritabilities: the diagonal of G over the diagonal of P. diag() pulls out
12 # the main diagonal of a square matrix -- the elements defined in section 1.2.
13 diag(G) / diag(P)
14 #> [1] 0.36 0.36
15
16 # Correlations: an off-diagonal over the root of the product of its diagonals.
17 G[1, 2] / sqrt(G[1, 1] * G[2, 2])  # genetic,      0.667
18 #> [1] 0.6666667
19 R[1, 2] / sqrt(R[1, 1] * R[2, 2])  # residual,     0.125
20 #> [1] 0.125
21 P[1, 2] / sqrt(P[1, 1] * P[2, 2])  # phenotypic,   0.32
22 #> [1] 0.32

```

Subtraction works the same way, and it is how $\mathbf{R}$ is usually obtained in practice: $\mathbf{R} = \mathbf{P} - \mathbf{G}$. Addition requires identical dimensions — there is no sensible answer to “what is the 2×2 plus the 3×3 ,” and R will refuse.

Scalar multiplication multiplies every element by the same number. It is how the relationship matrix becomes a covariance matrix: $\mathbf{G} = \mathbf{A}\sigma_a^2$ for a single trait, so for two full sibs with $\sigma_a^2 = 36$,

$$\mathbf{A}\sigma_a^2 = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \times 36 = \begin{bmatrix} 36 & 18 \\ 18 & 36 \end{bmatrix}$$

The variance ratio $\alpha = \sigma_e^2/\sigma_a^2$ is a scalar too, and it is the number that controls how much a mixed model shrinks its predictions. With the values above, $\alpha = 64/36 = 1.78$.

1.5 Matrix multiplication

Transpose, addition and scalar multiplication all leave the numbers where they are. Multiplication does not, and it is the operation everything else in the book is built on. It is also the one students find hardest, so we are going to do it before naming it.

1.5.1 Two totals

Flock A's two lambs weigh 48 and 52 kg; flock B's weigh 41 and 45. We want each flock's total weight. Here are the records, with a bookkeeping column marking which flock each lamb belongs to. Call that table **W**. Nothing is being modelled or estimated — we are adding up weights, and the ones and zeros are only there to say which lamb belongs in which total.

	flock A	flock B	y
lamb 1	1	0	48
lamb 2	1	0	52
lamb 3	0	1	41
lamb 4	0	1	45

Flock A's total: walk down the flock A column, pair each entry with the weight beside it, multiply, and add.

$$1(48) + 1(52) + 0(41) + 0(45) = 100$$

Flock B's total, the same walk down the other column:

$$0(48) + 0(52) + 1(41) + 1(45) = 86$$

Two totals, from two columns, against one list of weights. Write them as a pair:

$$\begin{bmatrix} 100 \\ 86 \end{bmatrix}$$

Two flock totals

$$\mathbf{W}' \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} \mathbf{y} \\ \begin{bmatrix} 48 \\ 52 \\ 41 \\ 45 \end{bmatrix} \end{matrix} = \begin{bmatrix} 100 \\ 86 \end{bmatrix}$$

$2 \times 4 \quad 4 \times 1 \quad 2 \times 1$

Two columns in, two totals out. That walk is matrix multiplication.

Figure 1.2: Row times column, one lamb at a time. The inner dimensions cancel: a 2×4 against a 4×1 gives a 2×1 . Animated in the web edition: <https://austin-putz.github.io/linear-models-in-animal-breeding>

What mattered and what did not. The numbers 48 and 52 were these lambs' weights — change the weights and the totals change. The number of columns we walked through was the number of flocks, and it fixed how many totals came out: two columns in, two totals out. But *the walk itself* — take a column, pair it

term by term with the list, multiply, add — never changed, and it would not change for 4 lambs or 4 million, for 2 flocks or 200. The one thing the walk needed was that the column and the list be the same length, so the pairing never ran out on either side.

That walk is matrix multiplication. Stand the bookkeeping table on its side so its columns become rows — that is the transpose from §1.3 — and the pair of totals is $\mathbf{W}'\mathbf{y}$, whose i th element pairs row i of $\mathbf{W}'$ with $\mathbf{y}$ term by term and sums:

$$(\mathbf{W}'\mathbf{y})_i = \sum_k (\mathbf{W}')_{ik} y_k$$

In general, for $\mathbf{A}$ of size $m \times n$ and $\mathbf{B}$ of size $n \times p$, the product $\mathbf{C} = \mathbf{AB}$ is $m \times p$ with

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Element c_{ij} is row i of the left matrix walked against column j of the right one. The condition for this to work is exactly the one we relied on without saying so: **the inner dimensions must match**. An $m \times n$ times an $n \times p$ works and gives $m \times p$; the two n 's meet in the middle and cancel, and what survives is the outside pair.

Check Yourself

You just multiplied a 2×4 by a 4×1 and got a 2×1 . Before reading on: what comes out of a 4×2 times a 4×1 , and why?

Answer. Nothing — it cannot be formed. The inner dimensions are 2 and 4, and they do not match, so the pairing runs out after two terms with two weights left over. Written the other way, $\mathbf{W}'\mathbf{y}$ is 2×4 times 4×1 and works. This is why the bookkeeping table gets transposed before it meets $\mathbf{y}$, and not for any deeper reason.

Definition — Conformable

Two matrices are **conformable** for an operation when their dimensions permit it. For addition and subtraction that means identical dimensions. For multiplication it means something weaker, which §1.5 arrives at by doing it rather than by declaring it.

Why it matters. Non-conformability is the most common error a student makes in R, and it is the useful kind: the operation stops rather than returning a wrong number. When `%*%` throws “non-conformable arguments”, it is telling you that the two objects do not line up — usually because a matrix needs transposing, and occasionally because the model itself is wrong.

The special matrices below are named by where they appear in this book, because that is how you will meet them.

```

1 # The bookkeeping table of the walk above: one column per flock.
2 W <- cbind(flockA = c(1, 1, 0, 0),
3           flockB = c(0, 0, 1, 1))
4 W
5 #>      flockA flockB
6 #> [1,]      1      0
7 #> [2,]      1      0
8 #> [3,]      0      1
9 #> [4,]      0      1
10
11 # Transpose it, then multiply. The two flock totals we worked by hand: 100, 86.
```

```

12 t(W) %*% y
13 #>      [,1]
14 #> flockA  100
15 #> flockB   86
16
17 # W'W counts instead of summing: how many lambs are in each flock.
18 t(W) %*% W
19 #>      flockA flockB
20 #> flockA      2      0
21 #> flockB      0      2

```

💡 In R — %*% is multiplication; * is not

$\mathbf{A} * \mathbf{B}$ in R multiplies element by element and is the Hadamard product of §1.11. Matrix multiplication is $\mathbf{A} \%*\% \mathbf{B}$. The trap is that for two matrices of the same shape, $\mathbf{A} * \mathbf{B}$ runs without error and returns something plausible-looking, so this fails silently.

Once you have written $\mathbf{t}(\mathbf{X}) \%*\% \mathbf{X}$ a few times, use `crossprod(X)` instead — and `crossprod(X, y)` for $\mathbf{X}'\mathbf{y}$. They give the same answer and are faster, because they never build $\mathbf{X}'$ at all. There is also `tcrossprod(X)` for $\mathbf{X}\mathbf{X}'$. The rest of this book uses the `crossprod` forms.

```

1 all.equal(crossprod(W), t(W) %*% W)
2 #> [1] TRUE
3 all.equal(crossprod(W, y), t(W) %*% y)
4 #> [1] TRUE

```

1.5.2 The rule that catches everyone

Now something you cannot read off the page. Take two 2×2 matrices with no symmetry and nothing in common:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$$

Multiply them — you now know how — and transpose the result:

$$\mathbf{AB} = \begin{bmatrix} 4 & 8 \\ 6 & 0 \end{bmatrix} \quad (\mathbf{AB})' = \begin{bmatrix} 4 & 6 \\ 8 & 0 \end{bmatrix}$$

There are two obvious candidates for what $(\mathbf{AB})'$ ought to equal. Form both:

$$\mathbf{A}'\mathbf{B}' = \begin{bmatrix} 2 & 4 \\ 13 & 2 \end{bmatrix} \quad \mathbf{B}'\mathbf{A}' = \begin{bmatrix} 4 & 6 \\ 8 & 0 \end{bmatrix}$$

The second one matches. The first is not even close.

What mattered and what did not. The particular entries did not matter — nothing about 2, 1, 0 or 3 made the second candidate win. The dimensions did not matter either; had $\mathbf{A}$ been 3×5 and $\mathbf{B}$ 5×2 , $\mathbf{A}'\mathbf{B}'$ would have been 5×3 times 2×5 and could not have been formed at all. What mattered is that transposing swaps which index runs along which side, so the two factors have to swap as well for the ends still to meet.

❗ Key Equation — The reversal rules

$$(\mathbf{AB})' = \mathbf{B}'\mathbf{A}' \tag{1.1}$$

In words: transposing a product reverses the order of the factors. Watch for the shape of this rule again later — the reversal is not something special about the transpose.

```

1 # matrix() fills COLUMN by column, so read these down the columns, not across.
2 A <- matrix(c(2, 0,
3             1, 3), nrow = 2)
4 B <- matrix(c(1, 2,
5             4, 0), nrow = 2)
6
7 A
8 #>      [,1] [,2]
9 #> [1,]    2    1
10 #> [2,]    0    3
11
12 B
13 #>      [,1] [,2]
14 #> [1,]    1    4
15 #> [2,]    2    0
16
17 t(A %*% B)      # (AB)'
18 #>      [,1] [,2]
19 #> [1,]    4    6
20 #> [2,]    8    0
21
22 t(B) %*% t(A)   # B'A' -- the same
23 #>      [,1] [,2]
24 #> [1,]    4    6
25 #> [2,]    8    0
26
27 t(A) %*% t(B)   # A'B' -- not the same
28 #>      [,1] [,2]
29 #> [1,]    2    4
30 #> [2,]   13    2

```

⚠ Common Mistake — assuming AB and BA are the same thing

They are usually not, and often only one of them can even be formed. With the same $\mathbf{A}$ and $\mathbf{B}$:

$$\mathbf{AB} = \begin{bmatrix} 4 & 8 \\ 6 & 0 \end{bmatrix} \quad \mathbf{BA} = \begin{bmatrix} 2 & 13 \\ 4 & 2 \end{bmatrix}$$

Same two matrices, different products. Matrix multiplication is **associative** ($(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$) and **distributive** ($\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$), so you may re-bracket freely — but you may never reorder. When a derivation in a later chapter moves a matrix across another one, it is doing something, and it is worth checking what.

This is the single most productive source of error in the chapters that follow. When you are assembling the mixed model equations and you need the block above the diagonal, it is $\mathbf{X}'\mathbf{Z}$, and the block below it is $\mathbf{Z}'\mathbf{X}$ — which is $(\mathbf{X}'\mathbf{Z})'$, not $\mathbf{X}'\mathbf{Z}$ again. Having formed $\mathbf{A}'\mathbf{B}'$ once by hand and watched it come out wrong is worth more than reading the rule three times.

1.6 Special matrices

Some matrices turn up so often that they have names. They are worth meeting *after* the operations, because most of them are defined by how they behave under those operations rather than by what they look like.

Matrix	What it is	Where it shows up
Identity I	1s on the diagonal, 0s elsewhere. Multiplying by it changes nothing: $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$	the residual structure $\mathbf{R} = \mathbf{I}\sigma_e^2$; the “no relationships” model of Ch 3
Null 0	all zeros	the off-diagonal blocks joining unrelated families
J	all ones	the mean model; $\mathbf{J}/n$ as the averaging operator
Diagonal	non-zero only on the diagonal	weights, unequal residual variances, $\mathbf{D}$ in $\mathbf{A} = \mathbf{TDT}'$ (Ch 4)
Symmetric	$a_{ij} = a_{ji}$	every covariance matrix; $\mathbf{X}'\mathbf{X}$; the whole MME left-hand side
Triangular	zeros above (or below) the diagonal	$\mathbf{L}$ from Cholesky (§1.12); $\mathbf{T}$ from the pedigree (Ch 4)
Incidence	entries 0 or 1 only	$\mathbf{X}$ and $\mathbf{Z}$ — almost entirely zeros, which is why Ch 24 exists
Block / partitioned	built from smaller matrices as tiles	the MME itself (Ch 6)
Positive definite	a validity condition — see §1.13	what a covariance matrix must be to be a covariance matrix

```

1  diag(3)                                # the identity, 3 x 3
2  #>      [,1] [,2] [,3]
3  #> [1,]    1    0    0
4  #> [2,]    0    1    0
5  #> [3,]    0    0    1
6  diag(c(2, 5, 9))                      # a diagonal matrix from a vector
7  #>      [,1] [,2] [,3]
8  #> [1,]    2    0    0
9  #> [2,]    0    5    0
10 #> [3,]    0    0    9
11 matrix(1, nrow = 2, ncol = 3)          # J, all ones, 2 x 3
12 #>      [,1] [,2] [,3]
13 #> [1,]    1    1    1
14 #> [2,]    1    1    1
15
16 # Y is the measurement matrix from section 1.2.
17 nrow(Y)
18 #> [1] 4
19 ncol(Y)
20 #> [1] 2
21 t(Y)                                    # not square: transposing turns 4 x 2 into 2 x 4
22 #>      [,1] [,2] [,3] [,4]
23 #> [1,]   48   52   41   45
24 #> [2,]   25   28   22   25
25
26 # A matrix that is mostly zeros, stored sparsely: only the positions of the
27 # non-zeros are kept, and a dot is printed wherever a zero would be.
28 Matrix(diag(4), sparse = TRUE)
29 #> 4 x 4 diagonal matrix of class "ddiMatrix"
30 #>      [,1] [,2] [,3] [,4]
31 #> [1,]    1    .    .    .
32 #> [2,]    .    1    .    .
33 #> [3,]    .    .    1    .

```

34 #> [4,] . . . 1

i Beyond This Book — how $\mathbf{X}$ is really stored

The incidence matrices $\mathbf{X}$ and $\mathbf{Z}$ in the table above are the reason this matters. The $\mathbf{X}$ built in §1.7 is 4×2 and almost half zeros. In a national evaluation it is millions of rows by tens of thousands of columns and more than 99.99% zeros, and no one ever writes it out. Sparse storage — keeping only the positions of the non-zeros — is what makes such a matrix fit in memory at all, and it changes how every operation in this chapter is actually performed. Chapter 24 shows *why* dense methods fail at that scale; the mechanics of sparse storage and factorisation belong to the companion volume on large-scale computation.

One more piece of notation before we start: $\mathbf{X}'$ — read “X transpose”, or more often just “X prime” — is the subject of §1.3. Some texts write $\mathbf{X}^T$ for the same thing.

1.7 Why matrices: one model, many animals

Four lambs were weighed at a year old. Two are from flock A, two from flock B.

Table 1.1: Example A — four lambs, two flocks.

Lamb	Flock	Yearling weight (kg)
1	A	48
2	A	52
3	B	41
4	B	45

We think each lamb’s weight is a flock level plus whatever is left over for that individual. Written one lamb at a time, that is four separate equations. Take flock B as the baseline and let b_1 be flock B’s level and b_2 be however much flock A runs above it:

$$\begin{aligned} 48 &= b_1 + b_2 + e_1 \\ 52 &= b_1 + b_2 + e_2 \\ 41 &= b_1 + e_3 \\ 45 &= b_1 + e_4 \end{aligned}$$

Every equation has the same two unknowns. What differs between them is only *which* unknowns appear — a 1 or a 0 in front of each. Pull those coefficients out and write them as a table, one row per lamb, one column per unknown:

$$\begin{bmatrix} 48 \\ 52 \\ 41 \\ 45 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix}$$

Four rows, one per record. Two columns, one per thing being estimated: the first column is all ones because every lamb has flock B’s baseline in it, and the second column is 1 exactly when the lamb is in flock A. Call the list of weights $\mathbf{y}$, the table of ones and zeros $\mathbf{X}$, the two unknowns $\mathbf{b}$, and the leftovers $\mathbf{e}$:

$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e}$$

Nothing has been gained yet — those are the same four equations with the arithmetic hidden. What has been gained is that the statement no longer mentions four. Add a fifth lamb and $\mathbf{y}$ gets a fifth entry and $\mathbf{X}$ a fifth row; the equation is unchanged. Add 40,000 lambs and it is still unchanged. That invariance is the whole reason this book is written in matrices: **the model statement stops depending on how many animals you have**, and by Chapter 6 the same four symbols will carry 100,000 pigs without a character of difference.

```

1 lambs <- data.frame(lamb = 1:4,
2                     flock = c("A", "A", "B", "B"),
3                     wt     = c(48, 52, 41, 45))
4
5 lambs
6 #>   lamb flock wt
7 #> 1     1     A 48
8 #> 2     2     A 52
9 #> 3     3     B 41
10 #> 4     4     B 45
11
12 # y is just the column of weights.
13 y <- lambs$wt
14 y
15 #> [1] 48 52 41 45
16
17 # X, built exactly as it is written on the page: a column of ones for flock B's
18 # baseline, then a 1 wherever the lamb is in flock A.
19 baseline <- c(1, 1, 1, 1)
20 flockA <- c(1, 1, 0, 0)
21 X <- cbind(baseline, flockA)
22 X
23 #>      baseline flockA
24 #> [1,]         1      1
25 #> [2,]         1      1
26 #> [3,]         1      0
27 #> [4,]         1      0

```

With $\mathbf{X}$ and $\mathbf{y}$ in hand, the two products of §1.5 finally have their book meanings:

```

1 crossprod(X)           # X'X -- the table of counts
2 #>      baseline flockA
3 #> baseline         4      2
4 #> flockA          2      2
5 crossprod(X, y)        # X'y -- the table of sums
6 #>      [,1]
7 #> baseline    186
8 #> flockA      100
9
10 # Xb hands each record its fitted value. b is solved for in §1.9.
11 b_hat <- c(43, 7)
12 X %*% b_hat
13 #>      [,1]
14 #> [1,]    50
15 #> [2,]    50
16 #> [3,]    43
17 #> [4,]    43

```

R will also build $\mathbf{X}$ for you from the data frame. That is what you will use in practice, but it is worth seeing only after you have built one by hand, because the function chooses the parameterisation and you need to be able to check which one it chose.

```

1 # The same X, from a formula. Listing flock B first makes it the baseline,
2 # which is the parameterisation we wrote out above.
3 lambs$flock <- factor(lambs$flock, levels = c("B", "A"))
4 X_auto <- model.matrix(~ flock, data = lambs)
5 X_auto
6 #>      (Intercept) flockA
7 #> 1             1      1
8 #> 2             1      1
9 #> 3             1      0
10 #> 4             1      0
11 #> attr("assign")
12 #> [1] 0 1
13 #> attr("contrasts")
14 #> attr("contrasts")$flock
15 #> [1] "contr.treatment"
16
17 # Two cosmetic differences from the X we built by hand: the first column is
18 # called "(Intercept)" rather than "baseline", and model.matrix() attaches
19 # bookkeeping attributes of its own. Ignore both -- the numbers are identical.
20 all.equal(X, X_auto, check.attributes = FALSE)
21 #> [1] TRUE

```

1.7.1 $\mathbf{X'X}$ counts and $\mathbf{X'y}$ sums

Now form $\mathbf{X'X}$ for the $\mathbf{X}$ just built — the one whose first column is all ones and whose second column flags flock A — and form $\mathbf{X'y}$ from it. Row 1 of $\mathbf{X'}$ against column 1 of $\mathbf{X}$ is $1(1) + 1(1) + 1(1) + 1(1) = 4$. Row 1 against column 2 is $1(1) + 1(1) + 1(0) + 1(0) = 2$. Row 2 against column 2 is $1(1) + 1(1) + 0(0) + 0(0) = 2$.

$$\mathbf{X'X} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \quad \mathbf{X'y} = \begin{bmatrix} 186 \\ 100 \end{bmatrix}$$

Look at what those four numbers are. Every element of $\mathbf{X}$ is a 0 or a 1, so multiplying two of them together gives 1 only when both are 1 — and summing the products just *counts the rows where both columns had a 1*.

- **$\mathbf{X'X}$ is a table of counts.** The 4 is the number of records. The bottom-right 2 is the number of records in flock A. The off-diagonal 2 counts the records that have both a baseline and a flock A indicator — also 2, because every flock A record has both.
- **$\mathbf{X'y}$ is a table of sums.** 186 is the grand total of all four weights; 100 is the flock A total we computed above.

These are not facts stated about $\mathbf{X'X}$ after the fact. They are the reason it gets built. Every left-hand side you will meet in this book is a table of counts with variance ratios added to some of its diagonal, and every right-hand side is a table of sums. A student who owns this reads Chapter 6's mixed model equations without effort.

Two more products worth naming while we are here. $\mathbf{Xb}$ hands each record its fitted value — with the solution we reach in §1.9, $\mathbf{b} = (43, 7)'$, it gives $(50, 50, 43, 43)'$, flock A's lambs getting $43 + 7$ and flock B's getting 43. And $\mathbf{Zu}$, which arrives in Chapter 3, hands each record its own animal's breeding value in exactly the same way.

1.8 Rank, singularity, and the determinant

The parameterisation in §1.7 was a choice. A more natural-looking one gives every flock its own column and keeps the baseline as well:

$$\mathbf{X}_3 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Three columns: a baseline, a flock A indicator, a flock B indicator. Look at them. The second column plus the third column equals the first, in every row. The third column carries no information the other two lack — knowing a lamb is not in flock A already tells you it is in flock B.

Columns that can be built from other columns are **linearly dependent**, and $\mathbf{X}_3$ has three columns of which only two carry independent information. Build it and check the claim directly — column 2 plus column 3 really is column 1:

```

1 X3 <- cbind(baseline = 1,
2             flockA   = c(1, 1, 0, 0),
3             flockB   = c(0, 0, 1, 1))
4 X3
5 #>      baseline flockA flockB
6 #> [1,]         1      1      0
7 #> [2,]         1      1      0
8 #> [3,]         1      0      1
9 #> [4,]         1      0      1
10
11 X3[, "flockA"] + X3[, "flockB"]      # equals the baseline column, every row
12 #> [1] 1 1 1 1
13 all.equal(X3[, "flockA"] + X3[, "flockB"], X3[, "baseline"],
14           check.attributes = FALSE)
15 #> [1] TRUE

```

R will tell you the rank directly. `qr()` factorises the matrix and its `$rank` component counts the independent columns:

```

1 qr(X3)$rank      # 2 -- not 3, even though there are three columns
2 #> [1] 2
3 ncol(X3)         # 3 columns, so X3 is rank deficient
4 #> [1] 3
5
6 qr(X)$rank       # the two-column X of section 1.7 is full rank
7 #> [1] 2
8 ncol(X)
9 #> [1] 2

```

i Definition — Linear dependence and rank

A set of columns is **linearly dependent** when at least one of them can be written as a linear combination (§1.1) of the others — it adds no information the rest do not already carry. Columns that cannot be are **linearly independent**.

The **rank** of a matrix is the number of linearly independent columns it has. A matrix whose rank equals its number of columns is **full rank**; one whose rank is smaller is **rank deficient**. $\mathbf{X}_3$ has three columns and rank 2, so it is rank deficient.

Why it matters. Rank is what decides whether a model's effects can be estimated at all. In animal breeding the answer is routinely no, and rank deficiency is normal rather than exceptional — every model with a mean and a contemporary group effect has it. Chapter 2 is largely about what to do next, and §1.10 gives the tool.

Form $\mathbf{X}_3'\mathbf{X}_3$ and the dependence is inherited:

$$\mathbf{X}_3'\mathbf{X}_3 = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

Again column 2 plus column 3 equals column 1 — the dependence has been inherited:

```

1  XtX3 <- crossprod(X3)
2  XtX3
3  #>           baseline flockA flockB
4  #> baseline           4      2      2
5  #> flockA            2      2      0
6  #> flockB            2      0      2
7
8  XtX3[, 2] + XtX3[, 3]      # still equals column 1
9  #> baseline   flockA   flockB
10 #>           4      2      2
11 qr(XtX3)$rank              # 2, in a 3 x 3 matrix
12 #> [1] 2

```

i Definition — Singular matrix

A square matrix is **singular** when its columns are linearly dependent — equivalently, when its rank is less than its dimension, and when its determinant is zero. Those three statements say the same thing about the same matrix. A square matrix that is not singular is **non-singular**, or **full rank**.

A fourth equivalent statement is that a singular matrix has no inverse; §1.9 defines the inverse and shows why the two ideas are the same one.

The practical meaning is that the system of equations the matrix defines has **no unique solution**. Here you could add any constant to the baseline and subtract it from both flock effects without changing a single fitted value, so there is no single right answer to report.

Why it matters. This is the word your software will use when it refuses to fit a model, and it is worth being able to translate it into “two of the things I asked for cannot be told apart from each other”. The response is not to give up: §1.10 solves the system anyway and tells you which quantities survive. In Chapter 6 the animal block of the mixed model equations is deliberately made non-singular by adding $\mathbf{A}^{-1}\alpha$ to it, which is why the animal model can return a breeding value even for an animal with no record of its own.

1.8.1 Where this material comes from

Rank, linear dependence and the determinant are nineteenth-century mathematics with no single originating paper, so nothing is cited for them above. What *does* have a literature is the question this section leads into: what to do with a model whose $\mathbf{X}$ is not full rank.

Two results settle it, and both arrived surprisingly late. R. C. Bose gave the definition of an **estimable function** in the 1940s. Penrose (Penrose 1955) defined the generalized inverse for arbitrary matrices, and Rao (Rao 1962) showed what it does for least squares when the normal equations are singular.

Those dates are worth pausing on, because they fall *after* Henderson had begun the work this book teaches. Searle — who took his doctorate in animal breeding under Henderson at Cornell — put it plainly in a later interview (Wells 2009): his group had “not kept up with the concept of estimability propounded by R. C. Bose”, nor were they “aware of Penrose’s (1955) generalized inverse matrix which, as Rao (1962) demonstrated, clarified the whole business of solving least squares equations which are so often not of full rank.”

§1.10 is that clarification, and Chapter 2 makes estimability precise.

This is the normal case in animal breeding, not the exceptional one. A contemporary group effect plus an overall mean is exactly the structure above. So is a sex effect with a mean, and a breed effect with a mean. Chapter 2 is largely about what to do with it, and the answer — §1.10 — is not to prevent it but to solve it anyway and be careful about which quantities you then quote.

1.8.2 The determinant

The **determinant** is a single number computed from a square matrix, written $|\mathbf{A}|$ or $\det(\mathbf{A})$. For a 2×2 it is the difference of the two diagonal products:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

For the full-rank $\mathbf{X}'\mathbf{X}$ of §1.5, $\det = (4)(2) - (2)(2) = 4$. For a matrix with a redundant column the determinant is 0 — the 3×3 above has $\det = 0$, and so does the 2×2 $\begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix}$, whose determinant is $100(100) - 100(100) = 0$.

`det()` does all three:

```

1  XtX <- crossprod(X)
2  XtX
3  #>           baseline flockA
4  #> baseline           4         2
5  #> flockA            2         2
6  (XtX[1, 1] * XtX[2, 2]) - (XtX[1, 2] * XtX[2, 1]) # ad - bc, by hand
7  #> [1] 4
8  det(XtX) # the same, 4
9  #> [1] 4
10
11 det(XtX3) # 0 -- the rank-deficient one
12 #> [1] 0
```

Do not memorise the general formula; you will never compute a determinant larger than 3×3 by hand, and R does it in one call. Memorise the meaning instead, in its two forms:

- **For a design matrix**, $\det(\mathbf{X}'\mathbf{X}) = 0$ means a column is redundant and the effects are not separately estimable.
- **For a covariance matrix**, $\det(\mathbf{G}) = 0$ means a genetic correlation of exactly ± 1 — one trait carries no information the other lacks. The matrix $\begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix}$ says two traits with $\sigma_a^2 = 100$ each and $r_g = 1$: genetically, the same trait measured twice. REML will not converge on such a matrix, and Chapter 21 explains why.

Here is that covariance matrix — two traits with $\sigma_a^2 = 100$ each and a genetic correlation of exactly 1:

```

1  Gsing <- matrix(c(100, 100,
2                    100, 100), nrow = 2)
3  Gsing
4  #>      [,1] [,2]
5  #> [1,]  100  100
6  #> [2,]  100  100
7
8  det(Gsing) # 0
9  #> [1] 0
```

```

10 qr(Gsing)$rank      # 1, in a 2 x 2 matrix
11 #> [1] 1
12
13 # The genetic correlation this matrix implies:
14 Gsing[1, 2] / sqrt(Gsing[1, 1] * Gsing[2, 2]) # exactly 1
15 #> [1] 1

```

And this is what R does when you ask it to invert one. `solve()` computes an inverse — §1.9 is where that is explained — and here it is deliberately being asked for one that does not exist:

```

1 solve(Gsing)
2 #> Error in `solve.default()`:
3 #> ! Lapack routine dgesv: system is exactly singular: U[2,2] = 0

```

That message is the one to recognise. “System is exactly singular” is R telling you a column of your matrix is redundant. When it appears in a real analysis it almost always means two effects in the model cannot be told apart from each other — not that anything is broken.

1.9 Inversion, solving, and partitioned matrices

1.9.1 Finding an inverse without a formula

The **inverse** of a square matrix $\mathbf{A}$ is the matrix $\mathbf{A}^{-1}$ for which $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. That definition is not just a description — it is enough to *find* one.

Take $\mathbf{X}'\mathbf{X} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$ from §1.5, call the unknown inverse $\begin{bmatrix} w & x \\ z & v \end{bmatrix}$, and demand that the product be the identity:

$$\begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} w & x \\ z & v \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiplying out, one element of the identity at a time, gives four equations in four unknowns:

$$\begin{aligned} 4w + 2z &= 1 & 4x + 2v &= 0 \\ 2w + 2z &= 0 & 2x + 2v &= 1 \end{aligned}$$

The left pair: subtract the second from the first to get $2w = 1$, so $w = 0.5$, and then $z = -0.5$. The right pair: subtract to get $2x = -1$, so $x = -0.5$, and then $v = 1$.

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

Multiply back to confirm: $\begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. It is the inverse.

What mattered and what did not. The particular entries 4, 2, 2, 2 set the answers, obviously. But look at the shape of what came out. Every entry is something over 4 — and 4 is $\det(\mathbf{X}'\mathbf{X})$, computed in §1.8. The two diagonal entries traded places (the 4 and the 2 in the original became a $2/4$ and a $4/4$). The two off-diagonal entries kept their positions but changed sign. None of that depended on the numbers; solve those four equations with any a , b , c , d and the same three things happen.

! Key Equation — The inverse of a 2×2

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (1.2)$$

In words: swap the diagonal entries, flip the sign of the off-diagonal ones, and divide everything by the determinant. In general $\mathbf{A}^{-1} = \text{adj}(\mathbf{A})/\det(\mathbf{A})$, of which this is the 2×2 case — the only one you will ever compute by hand.

The division makes §1.8 obvious in hindsight. If $\det(\mathbf{A}) = 0$ there is nothing to divide by, and the inverse does not exist. “Singular” and “not invertible” are the same statement.

i Definition — Inverse

The **inverse** of a square matrix $\mathbf{A}$, written $\mathbf{A}^{-1}$, is the matrix for which $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. It is the matrix equivalent of a reciprocal, and it exists only when $\mathbf{A}$ is non-singular.

Why it matters. Inverting is how a matrix equation gets solved, and $\mathbf{A}^{-1}$ — the inverse of the relationship matrix — is the single object that makes the animal model practical (Chapter 4). Note the two cautions that follow: do not compute an inverse merely to solve a system (below), and remember that the inverse of a covariance matrix is a *precision* matrix, which is why relationships enter Henderson’s equations as information rather than as covariance.

1.9.2 Three inverses worth recognising

The inverse of a **diagonal** matrix is the diagonal matrix of reciprocals — nothing else to do:

$$\begin{bmatrix} 4 & 0 \\ 0 & 10 \end{bmatrix}^{-1} = \begin{bmatrix} 0.25 & 0 \\ 0 & 0.1 \end{bmatrix}$$

The inverse of a **product** also has a rule, and you can predict it. Look at Equation 1.1, then write down what you think $(\mathbf{AB})^{-1}$ equals before reading the next sentence. — It is $\mathbf{B}^{-1}\mathbf{A}^{-1}$, and for the same reason: the ends have to keep meeting.

The inverse of a **covariance** matrix is a **precision** matrix, and it is worth pausing on. A covariance matrix says how much things vary together; its inverse says how much information you have about each after accounting for the others. This is the first hint of something that looks strange in Chapter 6: it is $\mathbf{A}^{-1}$, not $\mathbf{A}$, that appears in the mixed model equations — and Henderson’s rules for writing it down straight from a pedigree, without ever forming $\mathbf{A}$ at all (Henderson 1976), are what made the animal model computable. Relationships enter the equations as information, not as covariance.

1.9.3 Do not invert to solve

We want $\mathbf{b}$ from $\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{X}'\mathbf{y}$. Having the inverse, that is one multiplication:

$$\hat{\mathbf{b}} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \end{bmatrix} \begin{bmatrix} 186 \\ 100 \end{bmatrix} = \begin{bmatrix} 0.5(186) - 0.5(100) \\ -0.5(186) + 1(100) \end{bmatrix} = \begin{bmatrix} 43 \\ 7 \end{bmatrix}$$

Flock B’s mean is 43 kg and flock A runs 7 kg above it — which matches the raw means, 50 and 43, as it must for a design this balanced.

But do not compute an inverse in order to solve a system. `solve(A, b)` factorises and back-substitutes; `solve(A) %*% b` forms the entire inverse first and then multiplies. The two agree to rounding, but the second is slower, uses more memory, and is less numerically accurate, and the gap grows with size. Chapter 24 makes this quantitative. Form an inverse only when you want the inverse itself — which does

happen, because the diagonal of $(\mathbf{X}'\mathbf{X})^{-1}$ carries the standard errors, and in Chapter 6 the diagonal of the inverted coefficient matrix carries prediction error variance and accuracy. When you do need it for a symmetric matrix, `chol2inv(chol(A))` is the cheaper route: it inverts from the Cholesky factor of §1.12 rather than starting over.

Both give 43 and 7 here. The difference shows up at scale, so time them on a 500 x 500 system:

```

1  set.seed(2024)
2  big_X <- matrix(rnorm(600 * 500), nrow = 600, ncol = 500)
3  big_A <- crossprod(big_X)          # 500 x 500, symmetric, full rank
4  big_b <- rnorm(500)
5
6  time_solve <- system.time(solve(big_A, big_b))
7  time_invert <- system.time(solve(big_A) %*% big_b)
8
9  time_solve[["elapsed"]]             # solve(A, b)
10 #> [1] 0.005
11 time_invert[["elapsed"]]           # solve(A) %*% b
12 #> [1] 0.012

```

1.9.4 Partitioned matrices

The mixed model equations are a 2×2 system of *blocks*, not of scalars, so it is worth seeing that the 2×2 inverse formula works one level up. Partition the very matrix we just inverted, taking each block to be 1×1 :

$$\mathbf{X}'\mathbf{X} = \left[\begin{array}{c|c} 4 & 2 \\ \hline 2 & 2 \end{array} \right] = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix}$$

The key quantity is the **Schur complement** of $\mathbf{C}_{11}$ — what is left of $\mathbf{C}_{22}$ once $\mathbf{C}_{11}$'s contribution has been taken out of it:

$$\mathbf{S} = \mathbf{C}_{22} - \mathbf{C}_{21}\mathbf{C}_{11}^{-1}\mathbf{C}_{12} = 2 - 2\left(\frac{1}{4}\right)2 = 1$$

And $\mathbf{S}^{-1} = 1$ is exactly the bottom-right entry of the inverse we found by hand. The other blocks follow: $-\mathbf{C}_{11}^{-1}\mathbf{C}_{12}\mathbf{S}^{-1} = -\frac{1}{4}(2)(1) = -0.5$ off the diagonal, and $\mathbf{C}_{11}^{-1} + \mathbf{C}_{11}^{-1}\mathbf{C}_{12}\mathbf{S}^{-1}\mathbf{C}_{21}\mathbf{C}_{11}^{-1} = 0.25 + 0.25 = 0.5$ top-left. The same inverse, reached a different way.

Nothing in that argument used the fact that the blocks were 1×1 . When Chapter 7 absorbs the animal equations into the fixed-effect equations, and when Chapter 2 absorbs contemporary groups, they are computing a Schur complement.

i Derivation — the inverse of a partitioned matrix (optional)

Skip this on a first read. You already have the result above, verified on numbers; this shows it holds for blocks of any size.

Write $\mathbf{C} = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix}$ with $\mathbf{C}_{11}$ square and invertible, and factor it into a lower-triangular, a block-diagonal, and an upper-triangular piece:

$$\mathbf{C} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{C}_{21}\mathbf{C}_{11}^{-1} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{C}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{S} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{C}_{11}^{-1}\mathbf{C}_{12} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad \mathbf{S} = \mathbf{C}_{22} - \mathbf{C}_{21}\mathbf{C}_{11}^{-1}\mathbf{C}_{12}$$

Multiply the three out to confirm the factorisation. Now invert, using $(\mathbf{ABC})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$ from §1.9 — the reversal rule again — and noting that a unit-triangular block matrix inverts by flipping the sign of its one off-diagonal block:

$$\mathbf{C}^{-1} = \begin{bmatrix} \mathbf{C}_{11}^{-1} + \mathbf{C}_{11}^{-1}\mathbf{C}_{12}\mathbf{S}^{-1}\mathbf{C}_{21}\mathbf{C}_{11}^{-1} & -\mathbf{C}_{11}^{-1}\mathbf{C}_{12}\mathbf{S}^{-1} \\ -\mathbf{S}^{-1}\mathbf{C}_{21}\mathbf{C}_{11}^{-1} & \mathbf{S}^{-1} \end{bmatrix}$$

Setting every block to a scalar recovers Equation 1.2. Two consequences used later: the bottom-right block of the inverse depends on $\mathbf{C}_{22}$ only through the Schur complement, which is why prediction error variance for the animals accounts for the fixed effects having been estimated from the same data; and $\det(\mathbf{C}) = \det(\mathbf{C}_{11}) \det(\mathbf{S})$, which is how Chapter 21 evaluates a REML likelihood without forming the whole determinant.

```

1 XtX <- crossprod(X)
2 Xty <- crossprod(X, y)
3
4 solve(XtX)           # the inverse we found by hand
5 #>               baseline flockA
6 #> baseline      0.5   -0.5
7 #> flockA       -0.5    1.0
8 solve(XtX, Xty)      # b-hat = 43, 7 -- solve directly, the right way
9 #>               [,1]
10 #> baseline     43
11 #> flockA       7
12 solve(XtX) %*% Xty   # same answer, by forming the inverse first
13 #>               [,1]
14 #> baseline     43
15 #> flockA       7

```

1.10 Generalized inverses

The rank-deficient $\mathbf{X}_3'\mathbf{X}_3$ of §1.8 has no inverse, but it still has solutions — infinitely many of them, and a way of reaching one.

Definition — Generalized inverse

A **generalized inverse** of $\mathbf{A}$, written $\mathbf{A}^-$, is any matrix satisfying $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$. That single condition is enough to produce a solution to the system, and it is much weaker than the condition defining a true inverse — which is why a singular matrix has generalized inverses even though it has no inverse. Crucially, $\mathbf{A}^-$ is **not unique**. A singular matrix has infinitely many, and different ones give different solution vectors.

Why it matters. Every fixed-effect model in this book is solved with a generalized inverse, because every one of them is rank deficient. The consequence is the one thing to carry out of this section: the individual numbers in $\hat{\mathbf{b}}$ depend on which generalized inverse you happened to use, so they are not results. What you can report is defined next.

The easiest way to build one is to drop enough columns to leave a full-rank piece, invert that, and pad the result back out with zeros. There are three obvious choices here; take two of them.

Drop the baseline. Keep only the two flock columns, invert their 2×2 corner $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, and pad. The solution is

$$\hat{\mathbf{b}}_1 = (0, 50, 43)'$$

Built in R, the padding is what makes it a *generalized* inverse — the baseline's row and column stay zero:

```

1  XtX3 <- crossprod(X3)
2  Xty3 <- crossprod(X3, y)
3
4  Ginv_drop_baseline <- matrix(0, nrow = 3, ncol = 3)
5  Ginv_drop_baseline[2:3, 2:3] <- solve(XtX3[2:3, 2:3])
6  Ginv_drop_baseline
7  #>      [,1] [,2] [,3]
8  #> [1,]    0  0.0  0.0
9  #> [2,]    0  0.5  0.0
10 #> [3,]    0  0.0  0.5
11
12 drop(Ginv_drop_baseline %*% Xty3)      # 0, 50, 43
13 #> [1]  0 50 43

```

Drop the flock B column. Keep the baseline and flock A, which is exactly the $\mathbf{X}$ of §1.7, and pad the zero back in at the end:

$$\hat{\mathbf{b}}_2 = (43, 7, 0)'$$

```

1  Ginv_drop_flockB <- matrix(0, nrow = 3, ncol = 3)
2  Ginv_drop_flockB[1:2, 1:2] <- solve(XtX3[1:2, 1:2])
3
4  drop(Ginv_drop_flockB %*% Xty3)      # 43, 7, 0
5  #> [1] 43  7  0

```

Both really are generalized inverses. The defining property is $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$, and it holds for each:

```

1  all.equal(XtX3 %*% Ginv_drop_baseline %*% XtX3, XtX3)
2  #> [1] TRUE
3  all.equal(XtX3 %*% Ginv_drop_flockB %*% XtX3, XtX3)
4  #> [1] TRUE

```

These are not small disagreements. In the first, flock A's effect is 50 and the baseline is 0; in the second, flock A's effect is 7 and the baseline is 43. **Neither is more correct than the other**, and asking “what is the effect of flock A?” has no answer.

Now compute the fitted values from each. Both give $(50, 50, 43, 43)'$ — the two flock means, identically. And both give the same flock difference: $50 - 43 = 7$ from the first, $7 - 0 = 7$ from the second.

```

1  b1 <- Ginv_drop_baseline %*% Xty3
2  b2 <- Ginv_drop_flockB %*% Xty3
3
4  # Different solutions...
5  rbind(drop_baseline = drop(b1), drop_flockB = drop(b2))
6  #>      [,1] [,2] [,3]
7  #> drop_baseline    0   50   43
8  #> drop_flockB     43    7    0
9
10 # ...identical fitted values.
11 rbind(drop_baseline = drop(X3 %*% b1), drop_flockB = drop(X3 %*% b2))
12 #>      [,1] [,2] [,3] [,4]
13 #> drop_baseline   50   50   43   43
14 #> drop_flockB     50   50   43   43
15
16 # ...and an identical flock difference.
17 c(from_first = drop(b1)[2] - drop(b1)[3],
18    from_second = drop(b2)[2] - drop(b2)[3])

```

```

19 #> from_first from_second
20 #>           7           7

```

🔥 Check Yourself

Two different solution vectors, same fitted values. Which quantities can you report, and which can you not?

Answer. You can report anything that came out the same: the fitted values, the flock difference of 7 kg, the residuals, and any error variance computed from them. You cannot report the individual elements of $\hat{\mathbf{b}}$, because they changed. The general test is that a quantity is trustworthy when it is a **linear function of the data's expected values** — a combination $\mathbf{k}'\mathbf{b}$ for which $\mathbf{k}'$ can be written as some combination of rows of $\mathbf{X}$. Such a combination is called **estimable**, and Chapter 2 makes it precise. Here $(0, 1, -1)$ is estimable and $(0, 1, 0)$ is not.

i Definition — Estimable function

A linear combination $\mathbf{k}'\mathbf{b}$ of the effects is **estimable** when its value is the same no matter which generalized inverse was used to obtain $\hat{\mathbf{b}}$. The formal test is that $\mathbf{k}'$ must be expressible as a linear combination of the rows of $\mathbf{X}$.

Here the flock difference $(0, 1, -1)$ is estimable and comes to 7 kg every time; the individual flock A effect $(0, 1, 0)$ is not estimable and takes a different value under every solution.

Why it matters. Estimability is the line between a number you may publish and a number that is an artefact of your software's internal choices. Differences between levels, fitted values, and residuals are estimable; individual effects in a rank-deficient model are not. Chapter 2 makes the test precise, and Chapter 9 returns to it when genetic groups make the question harder.

Table 1.2: Three generalized inverses of the same $\mathbf{X}_3'\mathbf{X}_3$. The individual solutions disagree completely; everything a breeder would report agrees exactly.

Solution	$\hat{b}_\mu$	$\hat{b}_A$	$\hat{b}_B$	Fitted values	$\hat{b}_A - \hat{b}_B$
Drop the baseline	0	50	43	50, 50, 43, 43	7
Drop flock B	43	7	0	50, 50, 43, 43	7
MASS::ginv (Moore–Penrose)	31	19	12	50, 50, 43, 43	7

Table 1.2 collects all three. The third row is what R's `MASS::ginv()` produces — the Moore–Penrose inverse (Penrose 1955), which adds three further conditions to $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ and so is unique. It is a perfectly good g-inverse and it gives a third completely different answer, which is the point of the table. What `lm()` does is the second row: it silently drops a level and reports the rest relative to it.

```

1 # MASS ships with R, so nothing needs installing.
2 b3 <- MASS::ginv(XtX3) %*% Xty3
3
4 rbind(drop_baseline = drop(b1),
5       drop_flockB   = drop(b2),
6       moore_penrose = drop(b3))
7 #>           [,1] [,2] [,3]
8 #> drop_baseline    0  50  43
9 #> drop_flockB     43   7   0
10 #> moore_penrose    31  19  12

```

```

11
12 # Still the same fitted values, and still a 7 kg difference.
13 drop(X3 %*% b3)
14 #> [1] 50 50 43 43
15 drop(b3)[2] - drop(b3)[3]
16 #> [1] 7

```

R's own `lm()` takes the second route — it silently drops a level and reports everything relative to it. That is why an `lm()` summary shows one fewer coefficient than you have levels:

```

1 coef(lm(wt ~ flock, data = lambs))
2 #> (Intercept)      flockA
3 #>          43          7

```

1.11 Kronecker product, Hadamard product, and direct sum

Three products that are not matrix multiplication. Each builds a big matrix out of small ones, and each has exactly one job in this book.

1.11.1 Kronecker product — multiple traits

Two full sibs, two traits. The relationship matrix for the sibs and the genetic covariance matrix for the traits are both 2×2 :

$$\mathbf{A} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \quad \mathbf{G}_0 = \begin{bmatrix} 36 & 12 \\ 12 & 9 \end{bmatrix}$$

Two animals with two traits each means four breeding values, so their covariance matrix must be 4×4 . It is built by replacing each element of one matrix with that element times the whole of the other. Taking $\mathbf{G}_0$ as the outer matrix:

$$\mathbf{G}_0 \otimes \mathbf{A} = \left[\begin{array}{cc|cc} 36 & 18 & 12 & 6 \\ 18 & 36 & 6 & 12 \\ \hline 12 & 6 & 9 & 4.5 \\ 6 & 12 & 4.5 & 9 \end{array} \right]$$

In R the operator is `%x%`, and `kronecker()` is the same function spelled out:

```

1 # From here A is the relationship matrix, not the generic 2 x 2 of section 1.3.
2 A <- matrix(c(1, 0.5,
3             0.5, 1), nrow = 2)      # two full sibs
4 G0 <- matrix(c(36, 12,
5              12, 9), nrow = 2)      # the two traits
6
7 G0 %x% A                          # 4 x 4: animal within trait
8 #>      [,1] [,2] [,3] [,4]
9 #> [1,]   36   18  12.0  6.0
10 #> [2,]   18   36  6.0  12.0
11 #> [3,]   12    6  9.0  4.5
12 #> [4,]    6   12  4.5  9.0
13 dim(G0 %x% A)
14 #> [1] 4 4

```

Read it. The top-left block is $36\mathbf{A}$ — trait 1's genetic variance spread over the two sibs. The bottom-right is $9\mathbf{A}$, trait 2's. The off-diagonal blocks are $12\mathbf{A}$, holding every trait-1-with-trait-2 covariance: 12 on the

block's diagonal, which is one animal's two traits, and $12 \times 0.5 = 6$ off it, which is one animal's trait 1 with the *other* animal's trait 2 — halved, because sibs share half their additive genes.

The ordering matters and mixing the two up is the classic multi-trait error. Written as $\mathbf{G}_0 \otimes \mathbf{A}$, the rows run trait 1 for both animals, then trait 2 for both — **animal within trait**. Written the other way round it is **trait within animal**:

$$\mathbf{A} \otimes \mathbf{G}_0 = \left[\begin{array}{cc|cc} 36 & 12 & 18 & 6 \\ 12 & 9 & 6 & 4.5 \\ \hline 18 & 6 & 36 & 12 \\ 6 & 4.5 & 12 & 9 \end{array} \right]$$

```

1 A %x% G0          # the other ordering: trait within animal
2 #>      [,1] [,2] [,3] [,4]
3 #> [1,]   36 12.0   18  6.0
4 #> [2,]   12  9.0    6  4.5
5 #> [3,]   18  6.0   36 12.0
6 #> [4,]    6  4.5   12  9.0

```

Compare element (1, 2) between the two: 18 in the first, 12 in the second. Same sixteen numbers, different positions — which is exactly the error the box below warns about.

The general definition is that $\mathbf{A} \otimes \mathbf{B}$ replaces each a_{ij} with the block $a_{ij}\mathbf{B}$, so an $m \times n$ times a $p \times q$ gives $mp \times nq$. One property is worth carrying forward:

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$$

The consequence worth checking yourself: you can invert the two small factors instead of the big product, which is the difference between inverting two 2×2 s and one 4×4 — or, in Chapter 11, between a handful of traits and every animal in the pedigree.

```

1 all.equal(solve(G0 %x% A), solve(G0) %x% solve(A))
2 #> [1] TRUE

```

That mixed-product rule is what keeps multi-trait mixed model equations tractable — it lets you invert a Kronecker product by inverting its two small factors separately, since $(\mathbf{G}_0 \otimes \mathbf{A})^{-1} = \mathbf{G}_0^{-1} \otimes \mathbf{A}^{-1}$. Chapter 11 lives on this.

Common Mistake — trait within animal versus animal within trait

Both orderings are correct; they simply describe different arrangements of the same solution vector. The error is using one ordering for $\mathbf{G}_0 \otimes \mathbf{A}$ and the other for $\mathbf{Z}$ or for the solution vector, which silently scrambles the covariances rather than raising an error.

Fix the ordering once, write it down beside the model, and check it by picking one off-diagonal element and saying out loud which two breeding values it is the covariance of. In $\mathbf{G}_0 \otimes \mathbf{A}$ above, element (1, 4) is 6 — the covariance between animal 1's trait 1 and animal 2's trait 2. In $\mathbf{A} \otimes \mathbf{G}_0$ that same covariance sits at element (1, 4) as well, but for a different reason, and element (1, 2) has changed from 18 to 12.

1.11.2 Hadamard product $\#$ — non-additive effects

The Hadamard product multiplies element by element, so both matrices must be the same size and the result is that size too. For the two full sibs:

$$\mathbf{A} \# \mathbf{A} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \# \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.25 \\ 0.25 & 1 \end{bmatrix}$$

```

1 A * A      # element by element -- NOT matrix multiplication
2 #>      [,1] [,2]
3 #> [1,] 1.00 0.25
4 #> [2,] 0.25 1.00

```

Full sibs share half their additive genes but only a quarter of their additive-by-additive combinations, and the squaring is where that quarter comes from. $\mathbf{A}\#\mathbf{A}$ is the relationship matrix for additive $\times$ additive epistasis and $\mathbf{A}\#\mathbf{D}$ for additive $\times$ dominance; Chapter 17 uses both, and Chapter 16 builds the dominance matrix $\mathbf{D}$ they need.

1.11.3 Direct sum — independent blocks

The direct sum stacks matrices into a block-diagonal arrangement with zeros everywhere else:

$$\mathbf{A}_1 \oplus \mathbf{A}_2 = \begin{bmatrix} \mathbf{A}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 \end{bmatrix}$$

`Matrix::bdiag()` builds one. Two unrelated sib pairs have no covariance across the blocks, so the off-diagonal corners are zero:

```

1 as.matrix(bdiag(A, A))
2 #>      [,1] [,2] [,3] [,4]
3 #> [1,] 1.0 0.5 0.0 0.0
4 #> [2,] 0.5 1.0 0.0 0.0
5 #> [3,] 0.0 0.0 1.0 0.5
6 #> [4,] 0.0 0.0 0.5 1.0

```

This is what a set of unrelated families looks like, and what a residual covariance matrix looks like when animals are grouped into contemporary groups with no covariance across groups (Chapters 5 and 24).

1.12 Cholesky decomposition

Take the covariance matrix

$$\mathbf{M} = \begin{bmatrix} 100 & 60 \\ 60 & 100 \end{bmatrix}$$

Call this **Example C**. It is used again in §1.13.

and ask for a lower-triangular $\mathbf{L}$ with $\mathbf{M} = \mathbf{L}\mathbf{L}'$. Write the unknown $\mathbf{L}$ out and multiply:

$$\begin{bmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{bmatrix} \begin{bmatrix} \ell_{11} & \ell_{21} \\ 0 & \ell_{22} \end{bmatrix} = \begin{bmatrix} \ell_{11}^2 & \ell_{11}\ell_{21} \\ \ell_{11}\ell_{21} & \ell_{21}^2 + \ell_{22}^2 \end{bmatrix}$$

Now match entries against $\mathbf{M}$, in order, and watch what each step needs:

1. Top-left: $\ell_{11}^2 = 100$, so $\ell_{11} = 10$.
2. Below it: $\ell_{11}\ell_{21} = 60$. We already have $\ell_{11} = 10$, so $\ell_{21} = 6$.
3. Bottom-right: $\ell_{21}^2 + \ell_{22}^2 = 100$. We already have $\ell_{21} = 6$, so $\ell_{22}^2 = 100 - 36 = 64$ and $\ell_{22} = 8$.

$$\mathbf{L} = \begin{bmatrix} 10 & 0 \\ 6 & 8 \end{bmatrix}$$

What mattered and what did not. The numbers set the answers, but the *dependency* is the whole idea and it did not depend on the numbers at all: step 2 could not start until step 1 finished, and step 3 needed

step 2. That is why this is called a recursion rather than a formula. Written out for a matrix of any size, each entry uses only entries already computed:

$$\ell_{jj} = \sqrt{m_{jj} - \sum_{k=1}^{j-1} \ell_{jk}^2} \quad \ell_{ij} = \frac{1}{\ell_{jj}} \left(m_{ij} - \sum_{k=1}^{j-1} \ell_{ik} \ell_{jk} \right), \quad i > j$$

Note the square root. If any diagonal entry comes out as the square root of a negative number, the matrix is not positive definite and the factorisation fails — which makes Cholesky the standard *test* for whether a proposed covariance matrix is valid at all. That connects directly to §1.13.

Three uses, each named with its chapter:

1. **Solving.** Factor once, then solve two triangular systems by substitution — `chol()` followed by `backsolve()`. For a symmetric positive definite system such as $\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{X}'\mathbf{y}$ this is the numerically sound route, and it is roughly twice as fast as the general-purpose LU factorisation that plain `solve()` uses (Chapter 24).
2. **Simulating correlated breeding values.** If $\mathbf{z}$ is a vector of independent standard normals, then $\mathbf{a} = \mathbf{Lz}$ has covariance matrix $\mathbf{M}$ — because $\mathbf{L}\mathbf{L}' = \mathbf{L}\mathbf{L}' = \mathbf{M}$. Section 1.15 works out why that is the right calculation; for now, the R chunk below simply checks it on 10,000 samples. Every simulated two-trait animal in this book is generated this way (Chapters 11 and 22).
3. **Log-determinants.** $\log|\mathbf{M}| = 2 \sum_j \log \ell_{jj}$, which is how a REML likelihood gets evaluated without ever forming a determinant directly (Chapter 21).

A fourth connection, for later: Chapter 4 factors the relationship matrix as $\mathbf{A} = \mathbf{TDT}'$, which is a Cholesky-like factorisation whose triangular factor has a direct genetic reading — row i of $\mathbf{T}$ says what fraction of animal i 's genes came from each ancestor. That is where $\mathbf{A}^{-1}$'s famous sparsity comes from.

💡 In R — `chol()` returns the **upper** triangle

R's `chol()` returns the upper-triangular $\mathbf{R}$ with $\mathbf{M} = \mathbf{R}'\mathbf{R}$, not the lower-triangular $\mathbf{L}$ with $\mathbf{M} = \mathbf{L}\mathbf{L}'$ that the textbooks write. They are transposes of each other, so `L <- t(chol(M))`. Getting this wrong is a genuine trap because `chol(M) %*% t(chol(M))` runs without error and returns a symmetric matrix that is simply not $\mathbf{M}$. Simulating with `chol(G) %*% z` instead of `t(chol(G)) %*% z` gives breeding values with the wrong covariance structure and no warning at all. Check with `all.equal(L %*% t(L), M)` every time.

```

1 M <- matrix(c(100, 60,
2               60, 100), nrow = 2)
3
4 chol(M)           # R, UPPER triangular: 10 6 / 0 8
5 #>      [,1] [,2]
6 #> [1,]   10   6
7 #> [2,]    0   8
8 L <- t(chol(M))   # transpose it to get the L of the hand calculation
9 L
10 #>      [,1] [,2]
11 #> [1,]   10   0
12 #> [2,]    6   8
13
14 # Always check: LL' must give M back.
15 all.equal(L %*% t(L), M)
16 #> [1] TRUE

```

Now use $\mathbf{L}$ to simulate. Draw independent standard normals, multiply by $\mathbf{L}$, and the results carry $\mathbf{M}$'s covariance structure:

```

1  set.seed(2024)
2  z <- matrix(rnorm(2 * 10000), nrow = 2) # 10,000 pairs, independent
3  a <- L %%% z                             # now correlated
4
5  round(cov(t(a)), 1) # recovers M: 100, 60 / 60, 100
6  #>      [,1] [,2]
7  #> [1,] 103.0 61.5
8  #> [2,] 61.5 99.6

```

And the log-determinant, twice over — from the diagonal of $\mathbf{L}$, and directly:

```

1  2 * sum(log(diag(L)))
2  #> [1] 8.764053
3  log(det(M))
4  #> [1] 8.764053

```

1.13 Eigenvalues and eigenvectors

An **eigenvector** of a square matrix is a direction that the matrix does not rotate — it only stretches it. The stretch factor is the **eigenvalue**: $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$.

For Example C’s $\mathbf{M}$, try the direction $(1, 1)'$, meaning “both traits together”:

$$\begin{bmatrix} 100 & 60 \\ 60 & 100 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 160 \\ 160 \end{bmatrix} = 160 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Same direction, stretched by 160. Now try $(1, -1)'$, meaning “one trait up, the other down”:

$$\begin{bmatrix} 100 & 60 \\ 60 & 100 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 40 \\ -40 \end{bmatrix} = 40 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Same direction, stretched by 40. Those are the two eigenvalues, 160 and 40, and their eigenvectors are conventionally scaled to unit length as $(1, 1)'/\sqrt{2}$ and $(1, -1)'/\sqrt{2}$.

The interpretation is the reason to care. The two traits are positively correlated, so most of the variation is animals that are high for both or low for both — that is the 160 axis. Much less variation is animals high for one and low for the other — the 40 axis. Drawn as an ellipse, the eigenvectors are the axes and the square roots of the eigenvalues are their half-lengths.

```

1  # Folded: this is plot cosmetics, not algebra. The eigen-decomposition itself
2  # is in the visible chunk below.
3  eig <- eigen(M)
4  angle <- seq(0, 2 * pi, length.out = 400)
5  pts <- eig$vectors %%% diag(sqrt(eig$values)) %%% rbind(cos(angle), sin(angle))
6
7  par(mar = c(4, 4, 1, 1))
8  plot(pts[1, ], pts[2, ], type = "l", asp = 1, lwd = 2,
9        xlab = "trait 1 deviation", ylab = "trait 2 deviation")
10 abline(h = 0, v = 0, col = "grey80")
11 for (j in 1:2) {
12   axis_j <- eig$vectors[, j] * sqrt(eig$values[j])
13   arrows(0, 0, axis_j[1], axis_j[2], lwd = 2, length = 0.1,
14         col = c("firebrick", "steelblue")[j])
15   text(axis_j[1] * 1.18, axis_j[2] * 1.18,
16        bquote(lambda == .(eig$values[j])), cex = 0.9)
17 }

```

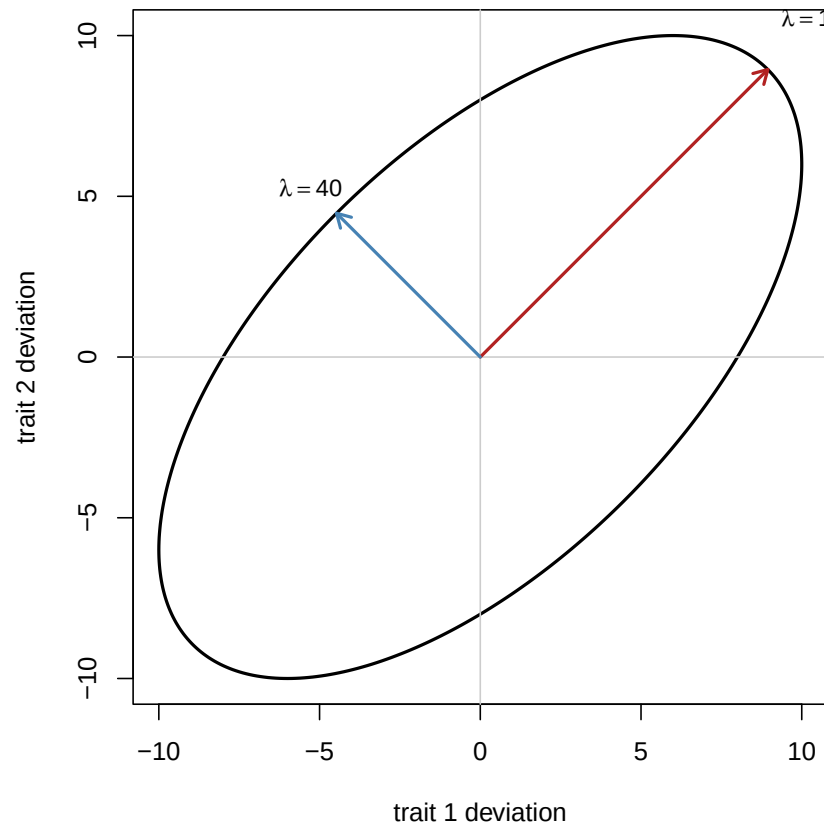


Figure 1.3: The covariance ellipse of Example C, with its eigenvectors drawn as axes. The long axis is the ‘both traits together’ direction (eigenvalue 160); the short axis is the contrast direction (eigenvalue 40).

🔥 Check Yourself

The eigenvalues of $\mathbf{M}$ are 160 and 40. Before computing it: what is $\det(\mathbf{M})$, and what would the determinant be if one eigenvalue were 0?

Answer. $160 \times 40 = 6400$, and directly, $100(100) - 60(60) = 10000 - 3600 = 6400$. The determinant is the product of the eigenvalues, so a single zero eigenvalue drives the whole determinant to zero — which is §1.8’s singularity seen from a different angle. Likewise the trace (the sum of the diagonal, $100 + 100 = 200$) equals the sum of the eigenvalues, $160 + 40 = 200$.

Three facts a breeder needs:

- **All eigenvalues > 0 means positive definite**, which is the condition for a matrix to be a valid covariance matrix. It is the same condition Cholesky tests in §1.12.

i Definition — Positive definite

A symmetric matrix $\mathbf{M}$ is **positive definite** when every one of its eigenvalues is greater than zero. Equivalently, $\mathbf{x}'\mathbf{M}\mathbf{x} > 0$ for every non-zero vector $\mathbf{x}$, and its Cholesky factorisation (§1.12) succeeds. If eigenvalues are zero but none are negative it is **positive semi-definite**.

Why it matters. A covariance matrix must be positive definite to be a covariance matrix at all. Since $\mathbf{x}'\mathbf{M}\mathbf{x}$ is the variance of a linear combination (§1.15), a non-positive-definite $\mathbf{G}$ would give some index of traits a *negative* variance, which is impossible. This is what your software means by “ $\mathbf{G}$ is not

positive definite” — a message you will meet in Chapter 21, and one that usually says the data cannot support as many traits as you asked for rather than that the program is broken.

- **A zero eigenvalue means a perfect genetic correlation.** Example C’s singular sibling $\begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix}$ has eigenvalues 200 and 0: all the variation is on the “together” axis and none on the contrast axis, because $r_g = 1$ and the two traits are genetically the same trait. REML cannot estimate such a matrix (Chapter 21).
- **Eigenvectors are the principal components of $\mathbf{G}$.** When a genetic covariance matrix for twenty traits has most of its variance in three eigenvalues, you can model three dimensions instead of twenty. That is what reduced-rank and factor-analytic random regression models do (Chapter 12).

i Derivation — why symmetric matrices behave so well (optional)

Skip this on a first read. It explains why every eigenvalue in this book is a real number and every pair of eigenvectors is at right angles — neither of which is true for matrices in general.

Real eigenvalues. Let $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{M}$ real and symmetric, allowing λ and $\mathbf{v}$ to be complex for the moment. Pre-multiply by the conjugate transpose $\bar{\mathbf{v}}'$:

$$\bar{\mathbf{v}}'\mathbf{M}\mathbf{v} = \lambda \bar{\mathbf{v}}'\mathbf{v}$$

The left side is its own conjugate transpose, because $\mathbf{M}$ is real and symmetric, so it is real. And $\bar{\mathbf{v}}'\mathbf{v} = \sum_i |v_i|^2$ is real and positive. A real number divided by a positive real number is real, so λ is real.

Orthogonal eigenvectors. Take two eigenpairs with $\lambda_1 \neq \lambda_2$. Then

$$\lambda_1 \mathbf{v}_2' \mathbf{v}_1 = \mathbf{v}_2' \mathbf{M} \mathbf{v}_1 = (\mathbf{M} \mathbf{v}_2)' \mathbf{v}_1 = \lambda_2 \mathbf{v}_2' \mathbf{v}_1$$

using $\mathbf{M}' = \mathbf{M}$ in the middle step. So $(\lambda_1 - \lambda_2) \mathbf{v}_2' \mathbf{v}_1 = 0$, and since the eigenvalues differ, $\mathbf{v}_2' \mathbf{v}_1 = 0$. The eigenvectors are perpendicular — which is why the ellipse above has axes at right angles, and why the eigen-decomposition $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}'$ has $\mathbf{V}^{-1} = \mathbf{V}'$.

Consequences used later. $\det(\mathbf{M}) = \prod_j \lambda_j$ and $\text{tr}(\mathbf{M}) = \sum_j \lambda_j$ both follow from that decomposition, and so does the statement that $\mathbf{M}$ is positive definite exactly when every $\lambda_j > 0$.

```

1 eig <- eigen(M)
2 eig$values      # 160 and 40
3 #> [1] 160  40
4 eig$vectors     # the sum axis and the contrast axis, scaled to length 1
5 #>      [,1]      [,2]
6 #> [1,] 0.7071068 -0.7071068
7 #> [2,] 0.7071068  0.7071068
8
9 # The determinant is the product of the eigenvalues.
10 det(M)
11 #> [1] 6400
12 prod(eig$values)
13 #> [1] 6400
14
15 # The trace is their sum.
16 sum(diag(M))
17 #> [1] 200
18 sum(eig$values)
19 #> [1] 200
20
21 # All eigenvalues positive means positive definite: a valid covariance matrix.
```

```

22 all(eig$values > 0)
23 #> [1] TRUE
24
25 # The rg = 1 matrix is not: one eigenvalue is exactly zero.
26 eigen(matrix(100, nrow = 2, ncol = 2))$values
27 #> [1] 200 0

```

A closely related tool, the **singular value decomposition**, factors any matrix — square or not — as $\mathbf{UDV}'$, and for a symmetric positive definite matrix it coincides with the eigen-decomposition. It is the workhorse behind numerically stable rank determination and behind the low-rank approximations used in large-scale evaluation. This book does not develop it; see the companion volume on computation.

1.14 Quadratic forms, traces, and matrix derivatives

This section exists to serve Chapter 21. Variance component estimation by REML is an exercise in differentiating a likelihood with respect to variance components, and the likelihood is built from quadratic forms and traces. Everything here is a tool you will pick up again there. Chapter 2 also needs the last part of it, three pages from now.

1.14.1 Quadratic forms

A **quadratic form** is $\mathbf{x}'\mathbf{A}\mathbf{x}$ — a row vector, a matrix, and the same vector again as a column. The result is always a single number. The simplest one in this book is the residual sum of squares. From §1.9, the residuals of Example A are

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\mathbf{b}} = \begin{bmatrix} 48 \\ 52 \\ 41 \\ 45 \end{bmatrix} - \begin{bmatrix} 50 \\ 50 \\ 43 \\ 43 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ -2 \\ 2 \end{bmatrix}$$

and

$$\mathbf{e}'\mathbf{e} = (-2)^2 + 2^2 + (-2)^2 + 2^2 = 16$$

```

1 b_hat <- drop(solve(crossprod(X), crossprod(X, y)))
2 e <- drop(y - X %*% b_hat)
3 e
4 #> [1] -2 2 -2 2
5
6 drop(crossprod(e))      # e'e = 16, the residual sum of squares
7 #> [1] 16

```

which is $\mathbf{e}'\mathbf{I}\mathbf{e}$, a quadratic form with $\mathbf{A} = \mathbf{I}$. Dividing by the residual degrees of freedom, $4 - 2 = 2$, gives $\hat{\sigma}_e^2 = 8 \text{ kg}^2$.

```

1 df <- nrow(X) - qr(X)$rank      # 4 records - 2 estimated effects
2 df
3 #> [1] 2
4 drop(crossprod(e)) / df        # sigma-hat squared = 8
5 #> [1] 8

```

The same construction gives $\mathbf{y}'\mathbf{y} = 8714$ as the uncorrected total sum of squares and $\hat{\mathbf{b}}'\mathbf{X}'\mathbf{y} = 43(186) + 7(100) = 8698$ as the part explained by the model, and $8714 - 8698 = 16$ recovers the residual sum of squares — the sums-of-squares partition Chapter 20 estimates variance components from.

```

1 yty      <- drop(crossprod(y))          # total
2 ss_model <- drop(b_hat %*% crossprod(X, y)) # explained by the model
3 c(total = yty, model = ss_model, residual = yty - ss_model)
4 #>      total      model residual
5 #>      8714      8698        16

```

1.14.2 Trace

The **trace** of a square matrix is the sum of its diagonal, $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$. It equals the sum of the eigenvalues (§1.13), and it has one property that gets used constantly in REML because it permits a reordering that nothing else does:

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$

There is no `trace()` in base R; you sum the diagonal yourself. Check the reordering property on two matrices whose products are plainly different:

```

1 AB <- A %*% GO
2 BA <- GO %*% A
3 AB                                     # not the same matrix
4 #>      [,1] [,2]
5 #> [1,]    42 16.5
6 #> [2,]    30 15.0
7 BA
8 #>      [,1] [,2]
9 #> [1,] 42.0  30
10 #> [2,] 16.5  15
11 c(trace_AB = sum(diag(AB)), trace_BA = sum(diag(BA))) # but the same trace
12 #> trace_AB trace_BA
13 #>      57      57

```

The result worth memorising concerns the **hat matrix** $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, which turns observations into fitted values: $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$. Its trace equals the rank of $\mathbf{X}$. For Example A, $\text{tr}(\mathbf{H}) = 2$, and the residual degrees of freedom are $n - \text{tr}(\mathbf{H}) = 4 - 2 = 2$ — the same 2 used above. **That is where degrees of freedom come from:** they count the dimensions the model consumed, and the trace of the projection matrix counts them whether or not $\mathbf{X}$ was full rank.

```

1 H <- X %*% solve(crossprod(X)) %*% t(X)
2
3 drop(H %*% y)          # the fitted values: 50 50 43 43
4 #> [1] 50 50 43 43
5 sum(diag(H))           # trace = 2
6 #> [1] 2
7 qr(X)$rank             # the same 2
8 #> [1] 2
9 nrow(X) - sum(diag(H)) # residual degrees of freedom
10 #> [1] 2

```

1.14.3 Matrix derivatives

Two derivatives, both of a scalar with respect to a vector. The result has the same shape as the vector you differentiated with respect to.

$$\frac{\partial(\mathbf{a}'\mathbf{x})}{\partial\mathbf{x}} = \mathbf{a} \quad \frac{\partial(\mathbf{x}'\mathbf{A}\mathbf{x})}{\partial\mathbf{x}} = 2\mathbf{A}\mathbf{x} \quad (\mathbf{A} \text{ symmetric})$$

The second is the matrix version of $d(ax^2)/dx = 2ax$, which is a good way to remember it. Apply them immediately. Least squares chooses $\mathbf{b}$ to minimise the residual sum of squares, which expands to

$$(\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{y}'\mathbf{y} - 2\mathbf{b}'\mathbf{X}'\mathbf{y} + \mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b}$$

Differentiate with respect to $\mathbf{b}$, using the first rule on the middle term and the second on the last (note $\mathbf{X}'\mathbf{X}$ is symmetric), and set the result to zero:

$$-2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{0} \quad \Rightarrow \quad \mathbf{X}'\mathbf{X}\hat{\mathbf{b}} = \mathbf{X}'\mathbf{y}$$

Those are the **normal equations**, and we have been solving them since §1.9 without having derived them. Chapter 2 opens with the result already in hand.

i Derivation — the two matrix derivatives (optional)

Skip this on a first read. Both rules are just ordinary calculus done one element at a time.

Linear form. $\mathbf{a}'\mathbf{x} = \sum_k a_k x_k$. Differentiating with respect to a single element x_i kills every term except the one containing it:

$$\frac{\partial}{\partial x_i} \sum_k a_k x_k = a_i$$

Stacking those partials into a vector gives $\mathbf{a}$.

Quadratic form. $\mathbf{x}'\mathbf{A}\mathbf{x} = \sum_j \sum_k x_j a_{jk} x_k$. Element x_i appears in the terms where $j = i$ and in the terms where $k = i$, so the product rule gives

$$\frac{\partial}{\partial x_i} \sum_j \sum_k x_j a_{jk} x_k = \sum_k a_{ik} x_k + \sum_j x_j a_{ji}$$

The first sum is row i of $\mathbf{A}\mathbf{x}$; the second is row i of $\mathbf{A}'\mathbf{x}$. In general the derivative is therefore $(\mathbf{A} + \mathbf{A}')\mathbf{x}$, and when $\mathbf{A}$ is symmetric the two sums are equal and it collapses to $2\mathbf{A}\mathbf{x}$. Every quadratic form in this book has a symmetric $\mathbf{A}$ — they are all covariance matrices or crossproducts — so the short version is the one to carry.

1.15 Variances of linear combinations: building V

Almost everything this book computes is a variance of something built out of something else, so this is the most-used result in it.

1.15.1 From one number to many

For a scalar, multiplying a random variable by a constant multiplies its variance by the square of that constant: $\text{Var}(ax) = a^2 \text{Var}(x)$. Doubling every breeding value quadruples the genetic variance.

Now the same thing with a matrix. Take three animals — a sire S and his two full-sib progeny, animals 1 and 2 — with additive relationship matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{bmatrix} \quad \mathbf{G} = \mathbf{A}\sigma_a^2 = 36\mathbf{A}$$

Only the two progeny have records, and each has one, so the matrix linking records to animals is

$$\mathbf{Z} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Two rows, one per record; three columns, one per animal in the pedigree. The sire's column is all zeros because he has no record of his own.

What is the covariance matrix of the two animals' breeding values *as they appear in the records*, $\text{Var}(\mathbf{Z}\mathbf{u})$? Work it out:

$$\mathbf{Z}\mathbf{G}\mathbf{Z}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 36 & 18 & 18 \\ 18 & 36 & 18 \\ 18 & 18 & 36 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 36 & 18 \\ 18 & 36 \end{bmatrix}$$

Z selected: pre-multiplying picked out the rows of $\mathbf{G}$ belonging to recorded animals, and post-multiplying by $\mathbf{Z}'$ picked out the matching columns. What survives is the sibs' 2×2 corner of $\mathbf{G}$ — the sire's row and column are gone, and the 18 off the diagonal is 0.5×36 , the genetic covariance between full sibs.

What mattered and what did not. The 36 was σ_a^2 and the 0.5 was the sib relationship — change either and every number changes. The dimensions came from two records and three animals. But the *arrangement* — pre-multiply by the matrix, post-multiply by its transpose — is what you would do for any $\mathbf{Z}$ at all, and it is the matrix version of squaring the constant.

! Key Equation — Variance of a linear combination

$$\text{Var}(\mathbf{A}\mathbf{y}) = \mathbf{A} \text{Var}(\mathbf{y}) \mathbf{A}' \quad (1.3)$$

In words: to get the variance of something built by multiplying, put the multiplier on the left, the original variance in the middle, and the transposed multiplier on the right. The transpose on the right is what keeps the answer square and symmetric.

Here $\mathbf{A}$ is **any matrix of constants** — this is the one place in the book where that letter is not the relationship matrix. In the worked example above the multiplier was $\mathbf{Z}$; below it is $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

1.15.2 Assembling V

Now add the residuals. Each record has residual variance $\sigma_e^2 = 64$, and residuals are independent of each other and of the breeding values, so $\mathbf{R} = \mathbf{I}\sigma_e^2$ and the two variances simply add:

$$\mathbf{V} = \text{Var}(\mathbf{y}) = \mathbf{Z}\mathbf{G}\mathbf{Z}' + \mathbf{R} = \begin{bmatrix} 36 & 18 \\ 18 & 36 \end{bmatrix} + \begin{bmatrix} 64 & 0 \\ 0 & 64 \end{bmatrix} = \begin{bmatrix} 100 & 18 \\ 18 & 100 \end{bmatrix}$$

Every number in $\mathbf{V}$ is something a breeder already knows. The diagonal is the phenotypic variance, 100 kg², so $h^2 = 36/100 = 0.36$ — Example B's trait, unchanged. The off-diagonal 18 is the phenotypic covariance between two full sibs, and $18/100 = 0.18$ is their phenotypic correlation, which is $\frac{1}{2}h^2$ exactly as sib-analysis theory says it should be.

That is the entire argument of the mixed model, in one matrix. Relationships live in $\mathbf{A}$; scaling by σ_a^2 makes them a covariance; $\mathbf{Z}$ propagates them from animals to records; adding $\mathbf{R}$ produces the covariance structure of the data. Chapter 6 does exactly this and then solves.

1.15.3 The variance of an estimate

The same rule gives standard errors. Since $\hat{\mathbf{b}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is a linear combination of $\mathbf{y}$, applying Equation 1.3 with $\text{Var}(\mathbf{y}) = \mathbf{I}\sigma_e^2$ and simplifying gives

$$\text{Var}(\hat{\mathbf{b}}) = (\mathbf{X}'\mathbf{X})^{-1}\sigma_e^2$$

For a contrast $\mathbf{k}'\hat{\mathbf{b}}$, one more application gives $\text{Var}(\mathbf{k}'\hat{\mathbf{b}}) = \mathbf{k}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{k}\sigma_e^2$. For Example A's flock difference, $\mathbf{k} = (0, 1)'$ picks out the bottom-right element of $(\mathbf{X}'\mathbf{X})^{-1}$, which is 1, so the variance is $1 \times 8 = 8$ and the standard error is $\sqrt{8} = 2.83$ kg. The 7 kg flock advantage is about 2.5 standard errors from zero — with four lambs, suggestive and no more.

In Chapter 6 the same diagonal, taken from the inverted mixed model coefficient matrix instead, becomes prediction error variance and then accuracy. It is the same calculation throughout.

```

1  # A is now the 3 x 3 relationship matrix: sire S, and his two full-sib progeny.
2  A <- matrix(c(1.0, 0.5, 0.5,
3               0.5, 1.0, 0.5,
4               0.5, 0.5, 1.0), nrow = 3, byrow = TRUE)
5
6  # Z links the two records to their animals. The sire's column is all zeros:
7  # he has no record of his own.
8  Z <- matrix(c(0, 1, 0,
9               0, 0, 1), nrow = 2, byrow = TRUE)
10
11 sigma2_a <- 36
12 sigma2_e <- 64
13
14 G <- A * sigma2_a          # relationships become a covariance matrix
15 R <- diag(sigma2_e, 2)     # residuals, independent
16
17 ZGZt <- Z %*% G %*% t(Z)  # Var(Zu): the sibs' corner of G
18 ZGZt
19 #>      [,1] [,2]
20 #> [1,]   36  18
21 #> [2,]   18  36
22
23 V <- ZGZt + R              # Var(y)
24 V
25 #>      [,1] [,2]
26 #> [1,]  100  18
27 #> [2,]   18 100
28
29 # Everything a breeder reads off V.
30 sigma2_a / V[1, 1]         # h2 = 0.36
31 #> [1] 0.36
32 V[1, 2] / V[1, 1]         # 0.18, the sibs' phenotypic correlation, = h2 / 2
33 #> [1] 0.18

```

The simulation is the part a student actually believes. Draw 50,000 sets of breeding values, add residuals, and the realised covariance of the records converges on $\mathbf{V}$:

```

1  set.seed(2024)
2  n_sim <- 50000
3
4  L <- t(chol(G))           # section 1.12
5  u <- L %*% matrix(rnorm(3 * n_sim), nrow = 3)  # breeding values for S, 1, 2
6  e <- matrix(rnorm(2 * n_sim, sd = sqrt(sigma2_e)), nrow = 2)
7  y_sim <- Z %*% u + e
8

```

```

9 round(cov(t(y_sim)), 1)      # converges on V: 100, 18 / 18, 100
10 #>      [,1] [,2]
11 #> [1,] 101.3 18.4
12 #> [2,] 18.4 99.5

```

And the same rule gives the standard error of the flock contrast from Example A:

```

1 k <- c(0, 1)                  # picks out the flock A advantage
2 XtX_inv <- solve(crossprod(X))
3 sigma2_e_hat <- 8              # from section 1.14
4
5 var_contrast <- drop(t(k) %*% XtX_inv %*% k) * sigma2_e_hat
6 var_contrast
7 #> [1] 8
8 sqrt(var_contrast)            # standard error, 2.83 kg
9 #> [1] 2.828427

```

1.16 Which section do you need for which chapter?

Nobody reads a matrix algebra chapter straight through and remembers it. Come back here.

Section	What it gives you	Needed by
1.1 Why linear algebra	what it is, and where a breeder meets it	orientation only
1.2 Matrices, vectors, scalars	dimensions, elements, notation, linear combination	all
1.3 Transpose	$\mathbf{A}'$; symmetry	all
1.4 Addition and scalars	$\mathbf{P} = \mathbf{G} + \mathbf{R}$; $\mathbf{G} = \mathbf{A}\sigma_a^2$; α	Ch 3, Ch 6, Ch 11
1.5 Multiplication	the row-by-column walk; $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$	Ch 2, Ch 6 — the MME itself
1.6 Special matrices	$\mathbf{I}$, diagonal, symmetric, triangular, block	all
1.7 One model, many animals	$\mathbf{y} = \mathbf{Xb} + \mathbf{e}$; $\mathbf{X}'\mathbf{X}$ as counts, $\mathbf{X}'\mathbf{y}$ as sums	Ch 2, and every chapter after
1.8 Rank and determinant	why breeding models are singular	Ch 2, Ch 5, Ch 21
1.9 Inversion and partitioning	$\mathbf{A}^{-1}$; Schur complement; do not invert to solve	Ch 4, Ch 6, Ch 7, Ch 24
1.10 Generalized inverses	non-unique solutions; estimable functions	Ch 2, Ch 9
1.11 Kronecker and Hadamard	$\mathbf{G}_0 \otimes \mathbf{A}$; $\mathbf{A} \# \mathbf{A}$	Ch 11, Ch 16, Ch 17
1.12 Cholesky	factorising, simulating, log-determinants	Ch 4, Ch 21, Ch 22, Ch 24
1.13 Eigenvalues	positive definiteness; principal components	Ch 12, Ch 21
1.14 Quadratic forms and traces	sums of squares; degrees of freedom; derivatives	Ch 2, Ch 20, Ch 21
1.15 Variance of a linear combination	$\mathbf{V} = \mathbf{ZGZ}' + \mathbf{R}$; PEV	Ch 3, Ch 6, Ch 11, Ch 23

Sections 1.2 to 1.5 are the ones to read in order. They build on each other — multiplication needs the transpose, and everything after §1.5 needs multiplication. From §1.6 onward you can enter wherever you need to.

If you are heading straight for the animal model in Chapter 6, the minimum is §1.5, §1.9 and §1.15 — but read §1.2 through §1.5 first if you have not done linear algebra before.

1.17 Further reading

Four sources, and when to reach for each.

(Searle 1982), *Matrix Algebra Useful for Statistics*. If you read one book alongside this chapter, read this one. Searle wrote it for statisticians rather than mathematicians, it covers every operation here in far more depth, and its treatment of rank, generalized inverses and quadratic forms is the natural next step from §1.8, §1.10 and §1.14. Searle took his doctorate in animal breeding under Henderson, which is part of why the examples feel familiar.

(Penrose 1955) and (Rao 1962). The two short papers behind §1.10 — Penrose defines the generalized inverse, Rao shows what it buys you when the normal equations are singular. Read them in that order, and only if you want to know where `MASS::ginv()` comes from.

(Henderson 1976). The paper that made the animal model practical, by giving rules to write $\mathbf{A}^{-1}$ directly from a pedigree without forming $\mathbf{A}$. You do not need it yet — Chapter 4 is built on it.

(Wells 2009). An interview rather than a result, and the most readable thing on this list: how matrix algebra, linear models and animal breeding grew up together at Cornell, told by someone who was there.

1.18 Key equations

Name	Equation	Section
The reversal rules	Equation 1.1	§1.5
The inverse of a 2×2	Equation 1.2	§1.9
Variance of a linear combination	Equation 1.3	§1.15

1.19 Exercises

Solutions are in Appendix H.

- Five ewes are weighed. Three are in flock 1, with weights 120, 124 and 131 kg; two are in flock 2, with weights 108 and 112 kg. Using a baseline plus a flock 1 indicator, write $\mathbf{X}$ and $\mathbf{y}$, then form $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{y}$ by hand. State in words what every element of each counts or sums. (*objective 1*)
- For two traits, $\mathbf{G} = \begin{bmatrix} 16 & 6 \\ 6 & 9 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} 48 & 2 \\ 2 & 16 \end{bmatrix}$. Form $\mathbf{P}$, then compute both heritabilities, the genetic correlation, and the phenotypic correlation. Which of the two traits would respond faster to selection, and why is that not the whole answer? (*objective 1*)
- Take the $\mathbf{X}'\mathbf{X}$ from exercise 1. Compute its determinant, invert it by hand using Equation 1.2, and solve for $\hat{\mathbf{b}}$. Verify with `solve()`, and confirm your two elements against the raw flock means. (*objective 1*)
- Six lambs in three flocks of two weigh 50 and 54; 44 and 48; 39 and 41 kg. Build the overparameterised $\mathbf{X}$ with a baseline and all three flock columns. Show it is rank deficient and give its rank. Then produce two different generalized-inverse solutions, show that the fitted values agree, and give one estimable function on which they agree. (*objective 2*)
- With $\mathbf{G}_0 = \begin{bmatrix} 36 & 12 \\ 12 & 9 \end{bmatrix}$ for two traits and $\mathbf{A}$ the relationship matrix of three animals — a sire and two of his half-sib progeny out of unrelated dams — form both $\mathbf{G}_0 \otimes \mathbf{A}$ and $\mathbf{A} \otimes \mathbf{G}_0$. In each, identify the element that is $\text{cov}(a_{1,\text{trait 1}}, a_{3,\text{trait 2}})$, and say which ordering each matrix uses. (*objective 3*)

6. Let $\mathbf{G} = \begin{bmatrix} 64 & 24 \\ 24 & 25 \end{bmatrix}$ be a genetic covariance matrix for two traits.
- Compute its Cholesky factor $\mathbf{L}$ by hand, entry by entry, and verify $\mathbf{LL}' = \mathbf{G}$. (*objective 4*)
 - In R, use $\mathbf{L}$ to simulate 10,000 pairs of correlated breeding values and confirm that the realised covariance matrix recovers $\mathbf{G}$. Remember that `chol()` returns the upper factor. (*objective 4*)
 - Those 10,000 animals are each measured once for trait 1, so $\mathbf{Z} = \mathbf{I}$ and $\mathbf{R} = \mathbf{I}\sigma_e^2$ with $\sigma_e^2 = 36$. Use Equation 1.3 to give $\text{Var}(\mathbf{Zu})$ and then $\mathbf{V}$ for trait 1, and state the heritability. (*objective 5*)

Chapter 2

Linear Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Write** a linear model in matrix form and build $\mathbf{X}$ for a given design
2. **Compare** parameterizations and show they give identical fitted values
3. **Solve** $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ by hand and reproduce it with `lm()`
4. **Explain** estimability and why individual solutions need not be unique
5. **Interpret** residuals and estimate σ^2_e

Assumes Ch 1.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Fixed effects only. Build $\mathbf{X}$, solve the normal equations, confront non-full-rank.

Chapter 3

Mixed Models Without Relationships

Learning Objectives

By the end of this chapter, you will be able to:

1. **Distinguish** fixed from random effects and state when each is appropriate
2. **Build $\mathbf{Z}$** for a random classification effect
3. **Assemble** and solve the MME with a variance ratio
4. **Contrast** BLUE of fixed effects with BLUP of random effects, and explain shrinkage
5. **Reproduce** the hand solution with `lmer()`

Assumes Ch 2.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Introduce $\mathbf{Z}$ and random effects with no genetics attached. $\mathbf{G} = \mathbf{I}$.

Chapter 4

Pedigrees and Relationship Matrices

Learning Objectives

By the end of this chapter, you will be able to:

1. **Store**, sort, and renumber a pedigree correctly
2. **Build $\mathbf{A}$** by the tabular method by hand
3. **Compute** inbreeding coefficients and interpret the diagonal of $\mathbf{A}$
4. **Build $\mathbf{A}^{-1}$** directly by Henderson's rules, with and without inbreeding
5. **Reproduce** both in R

Assumes Ch 1.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. $\mathbf{A}$ and $\mathbf{A}^{-1}$ — the object that makes the animal model work.

Chapter 5

Data Preparation, Contemporary Groups, and Connectedness

Learning Objectives

By the end of this chapter, you will be able to:

1. **Define** contemporary groups and justify a definition for a given trait
2. **Diagnose** and handle small and singleton contemporary groups
3. **Assess** genetic connectedness and explain the consequences of disconnection
4. **Detect** and resolve pedigree errors
5. **Apply** an editing pipeline and document its decisions reproducibly

Assumes Ch 2, Ch 4.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. What goes wrong before a model is ever fitted. This is where real analyses fail.

Part II

Part II: Core Animal Models

Chapter 6

The Animal Model

Learning Objectives

By the end of this chapter, you will be able to:

1. **Build** and solve the animal-model MME by hand for a small pedigree
2. **Explain** the role of σ^2_e/σ^2_a and what happens as it grows
3. **Obtain** PEV, accuracy, and reliability from the inverse of the LHS
4. **Decompose** an EBV into parent average, yield deviation, and progeny contribution
5. **Explain** BLUP as a selection index with simultaneously estimated fixed effects

Assumes Ch 2 (X), Ch 3 (Z), Ch 4 (A^{-1}).

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. The central chapter. Solve the full MME with $\mathbf{A}^{-1}$, then interpret what came out.

Chapter 7

Equivalent and Reduced Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Derive** the sire model and state the assumptions that make it an approximation
2. **Relate** sire EBVs to animal-model EBVs computed on the same data
3. **Build** the sire–maternal grandsire relationship matrix and its inverse
4. **Set** up a reduced animal model and back-solve for non-parents
5. **Judge** when a reduced model is worth the information it discards

Assumes Ch 6.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Sire, sire–MGS, and reduced animal models, framed as data reduction and model equivalence rather than as a step backward.

Chapter 8

Random Environmental Effects

Learning Objectives

By the end of this chapter, you will be able to:

1. **Distinguish** permanent from temporary environmental effects
2. **Define** repeatability and state its relationship to heritability
3. **Set** up and solve a repeatability model with multiple records per animal
4. **Model** a common environmental (litter) effect and explain its confounding with relationship
5. **Interpret** variance ratios when several random effects compete for the same variance

Assumes Ch 6.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Repeatability and common environmental effects — a second random effect that is not genetic.

Chapter 9

Genetic Groups

Learning Objectives

By the end of this chapter, you will be able to:

1. **Explain** what the base population assumption means and when it fails
2. **Decide** how to form unknown parent groups for a given population
3. **Set** up and solve an animal model with groups
4. **Interpret** group solutions and their effect on estimated genetic trend
5. **State** the limitations of UPG and what metafounders change

Assumes Ch 4, Ch 6.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Unknown parent groups, and an honest statement of what they get wrong.

Chapter 10

Maternal Effects Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Separate** direct and maternal genetic contributions to a phenotype
2. **Build** the MME with correlated direct and maternal genetic effects
3. **Include** a maternal permanent environmental effect and justify it
4. **Interpret** the direct–maternal genetic correlation and the controversy surrounding it
5. **Compute** total maternal and total genetic merit for selection

Assumes Ch 6, Ch 8.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Two correlated genetic effects on one phenotype.

Part III

Part III: Advanced Model Structures

Chapter 11

Multivariate Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Write** the multivariate MME using the Kronecker product
2. **Solve** a two-trait model by hand with equal design matrices
3. **Handle** missing records and unequal design matrices
4. **Explain** how genetic correlation moves information between traits
5. **Describe** how dimension reduction makes many-trait models tractable

Assumes Ch 6, Ch 1 (§1.7).

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Several traits at once; the Kronecker product earns its keep.

Chapter 12

Longitudinal Data and Random Regression

Learning Objectives

By the end of this chapter, you will be able to:

1. **Distinguish** fixed from random regression and state what each can model
2. **Build** a covariate matrix of Legendre polynomials
3. **Set** up and solve a random regression model
4. **Recover** the covariance function and per-time-point variances from RR coefficients
5. **Fit** and interpret a reaction norm as a special case

Assumes Ch 8, Ch 11.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Traits measured along a trajectory.

Chapter 13

Social Interaction Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Explain** how a group-mate's genes become part of an animal's phenotype
2. **Build** the direct and indirect genetic effect design matrices
3. **Define** the total breeding value and show why it, not the direct effect, is the selection criterion
4. **Explain** how group size and composition affect identifiability
5. **Recognize** when indirect genetic effects and a random group effect are confounded

Assumes Ch 3, Ch 6, Ch 11.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. An animal's phenotype depends on its group-mates' genes.

Part IV

Part IV: Categorical and Time-to-Event Traits

Chapter 14

Threshold Models for Categorical Traits

Learning Objectives

By the end of this chapter, you will be able to:

1. **Explain** the liability concept and the threshold model
2. **Set** up a threshold model for a binary trait and solve it iteratively
3. **Extend** to more than two ordered categories with multiple thresholds
4. **Convert** heritability between the observed and underlying scales
5. **Fit** a joint model for a categorical and a continuous trait

Assumes Ch 6, Ch 11.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Binary and ordered categorical traits in one chapter — same model, more thresholds.

Chapter 15

Survival Analysis

Learning Objectives

By the end of this chapter, you will be able to:

1. **Define** censoring and explain why it breaks a standard linear model
2. **Distinguish** true from functional longevity
3. **Contrast** a binary stayability model with a true time-to-event analysis
4. **Fit** a proportional hazards model containing a genetic effect
5. **Interpret** survival EBVs and relate them to economic merit

Assumes Ch 6, Ch 14.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Time-to-event data and censoring.

Part V

Part V: Non-additive Genetic Effects

Chapter 16

Dominance Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Distinguish** breeding value from genotypic value
2. **Build** the dominance relationship matrix **D** from a pedigree
3. **Set** up and solve an animal model with additive and dominance effects
4. **Explain** why dominance and common environment confound
5. **Use** predicted dominance for mate allocation

Assumes Ch 4, Ch 6.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Genotypic value, not just breeding value — and the one place non-additive effects change a decision.

Chapter 17

Epistasis Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Define** additive $\times$ additive, additive $\times$ dominance, and dominance $\times$ dominance variance
2. **Build** epistatic relationship matrices as Hadamard products of **A** and **D**
3. **Set** up a model containing an epistatic term
4. **Explain** why epistatic variance is nearly unidentifiable from pedigree data
5. **Justify** the additive model used throughout the rest of the book

Assumes Ch 1 (§1.7), Ch 16.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Capstone to Part V, answering the question the whole book raises: why is everything else additive?

Part VI

Part VI: Multibreed and Crossbred Evaluation

Chapter 18

Multibreed Models

Learning Objectives

By the end of this chapter, you will be able to:

1. **Explain** why one **A** and one variance are wrong across breeds
2. **Model** breed composition and heterosis as covariates
3. **Define** segregation variance and state when it matters
4. **Build** a breed-specific covariance structure
5. **Explain** metafounders as a generalization of unknown parent groups

Assumes Ch 9, Ch 11.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. More than one breed in a single evaluation.

Chapter 19

Crossbred Evaluation and CCPS

Learning Objectives

By the end of this chapter, you will be able to:

1. **Explain** why purebred performance predicts crossbred performance imperfectly
2. **Define** the purebred–crossbred genetic correlation r_{pc} and its consequences
3. **Set** up purebred and crossbred performance as two correlated traits
4. **Judge** when crossbred data is worth collecting
5. **Describe** how genomic breed-origin methods extend this

Assumes Ch 11, Ch 18.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Selecting purebreds for crossbred performance.

Part VII

Part VII: Variance Component Estimation

Chapter 20

ANOVA and Henderson's Methods

Learning Objectives

By the end of this chapter, you will be able to:

1. **Estimate** variance components from a balanced half-sib design by ANOVA
2. **Compute** heritability and its standard error by hand
3. **Describe** Henderson's Methods I, II, and III and their scopes
4. **Show** how unbalanced data breaks the ANOVA approach
5. **Explain** why these methods were superseded

Assumes Ch 3, Ch 7.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Where variance components came from before REML, and why they had to be replaced.

Chapter 21

REML

Learning Objectives

By the end of this chapter, you will be able to:

1. **State** the difference between ML and REML and why REML is preferred
2. **Describe** the EM, Newton–Raphson, and average information algorithms
3. **Run** AI-REML and interpret its output
4. **Obtain** standard errors of variance components and of heritability
5. **Diagnose** convergence failure in multi-trait analyses

Assumes Ch 20.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. The workhorse.

Chapter 22

Bayesian Estimation via Gibbs Sampling

Learning Objectives

By the end of this chapter, you will be able to:

1. **State** the Bayesian formulation of the animal model and its priors
2. **Derive** the full conditional distributions
3. **Implement** a Gibbs sampler for a single-trait animal model
4. **Assess** convergence and mixing
5. **Summarize** posteriors and compare them to REML point estimates

Assumes Ch 21.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. The third route to the same variance components, and the one that gives full distributions.

Part VIII

Part VIII: Validation and Computation

Chapter 23

Validating Genetic Evaluations

Learning Objectives

By the end of this chapter, you will be able to:

1. **Distinguish** accuracy, bias, and dispersion as separate failure modes
2. **Design** a forward (time-split) validation
3. **Apply** the LR method and interpret its statistics
4. **Perform** cross-validation appropriately for related individuals
5. **Diagnose** the cause of a failed validation

Assumes Ch 6, Ch 21.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. How do you know the evaluation works?

Chapter 24

Solving the Mixed Model Equations at Scale

Learning Objectives

By the end of this chapter, you will be able to:

1. **Explain** why direct inversion is infeasible for a national evaluation
2. **Describe** the sparsity of the MME and of $\mathbf{A}^{-1}$
3. **Implement** Jacobi and Gauss–Seidel iteration on a small system
4. **Describe** PCG and iteration on data conceptually
5. **Explain** how reliabilities are approximated when the LHS cannot be inverted

Assumes Ch 6.

Chapter in progress

This chapter has not been written yet. The objectives above are the contract it will be written against — they are taken from `CHAPTERS.md`, which is the outline for the whole book.

Purpose. Expose students to what production software actually does. Deliberately thin.

Part IX

Appendices

Chapter 25

Notation reference

 Appendix in progress

This appendix has not been written yet.

Contents. Authoritative symbol list; check before introducing any new symbol

Chapter 26

Derivation of BLUP and the MME

 Appendix in progress

This appendix has not been written yet.

Contents. Henderson's derivation; proof that $\hat{\mathbf{b}}$ is GLS and $\hat{\mathbf{a}}$ is BLUP

Chapter 27

Fast inbreeding computation

 Appendix in progress

This appendix has not been written yet.

Contents. Meuwissen & Luo algorithm, worked

Chapter 28

Legendre polynomials

 Appendix in progress

This appendix has not been written yet.

Contents. Constructing Φ at given ages or time points

Chapter 29

Deregression of breeding values

 Appendix in progress

This appendix has not been written yet.

Contents. Procedure and worked example

Chapter 30

R code reference

 Appendix in progress

This appendix has not been written yet.

Contents. Common operations, indexed by task

Chapter 31

Datasets

 Appendix in progress

This appendix has not been written yet.

Contents. Every dataset used, with structure and provenance

Chapter 32

Solutions to exercises

 Appendix in progress

This appendix has not been written yet.

Contents. All chapters

Chapter 33

BLUPF90 guide

 Appendix in progress

This appendix has not been written yet.

Contents. Parameter files, running, reading output

References

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- Henderson, C. R. 1975. “Best Linear Unbiased Estimation and Prediction Under a Selection Model.” *Biometrics* 31 (2): 423–47. <https://doi.org/10.2307/2529430>.
- Henderson, C. R. 1976. “A Simple Method for Computing the Inverse of a Numerator Relationship Matrix Used in Prediction of Breeding Values.” *Biometrics* 32 (1): 69–83. <https://doi.org/10.2307/2529339>.
- Patterson, H. D., and R. Thompson. 1971. “Recovery of Inter-Block Information When Block Sizes Are Unequal.” *Biometrika* 58 (3): 545–54. <https://doi.org/10.1093/biomet/58.3.545>.
- Penrose, Roger. 1955. “A Generalized Inverse for Matrices.” *Mathematical Proceedings of the Cambridge Philosophical Society* 51 (3): 406–13. <https://doi.org/10.1017/S0305004100030401>.
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- Searle, Shayle R. 1982. *Matrix Algebra Useful for Statistics*. John Wiley & Sons.
- Wells, Martin T. 2009. “A Conversation with Shayle r. Searle.” *Statistical Science* 24 (2): 244–54. <https://doi.org/10.1214/08-STS259>.

Additional Resources

Textbooks

- **Henderson, C. R. (1984).** *Applications of Linear Models in Animal Breeding*. University of Guelph.
- **Lynch, M., & Walsh, B. (1998).** *Genetics and Analysis of Quantitative Traits*. Sinauer Associates.
- **Mrode, R. A. (2014).** *Linear Models for the Prediction of Animal Breeding Values* (3rd ed.). CABI.
- **Falconer, D. S., & Mackay, T. F. C. (1996).** *Introduction to Quantitative Genetics* (4th ed.). Longman.

Key Papers

Mixed Model Theory

- **Henderson, C. R. (1975).** Best linear unbiased estimation and prediction under a selection model. *Biometrics*, 31(2), 423-447.
- **Robinson, G. K. (1991).** That BLUP is a good thing: The estimation of random effects. *Statistical Science*, 6(1), 15-32.

- **Patterson, H. D., & Thompson, R. (1971).** Recovery of inter-block information when block sizes are unequal. *Biometrika*, 58(3), 545-554.

Non-additive Effects (Ch 16-17)

- **Henderson, C. R. (1985).** Best linear unbiased prediction of nonadditive genetic merits in noninbred populations. *Journal of Animal Science*, 60(1), 111-117. <https://doi.org/10.2527/jas1985.601111x>
- **Hill, W. G., Goddard, M. E., & Visscher, P. M. (2008).** Data and theory point to mainly additive genetic variance for complex traits. *PLoS Genetics*, 4(2), e1000008. <https://doi.org/10.1371/journal.pgen.1000008>
- **Hivert, V., Sidorenko, J., Rohart, F., et al. (2021).** Estimation of non-additive genetic variance in human complex traits from a large sample of unrelated individuals. *The American Journal of Human Genetics*, 108(5), 786-798. <https://doi.org/10.1016/j.ajhg.2021.02.014>

i Reference list is incomplete

This page currently holds only references verified against the literature. Chapter-specific references are added as each chapter is drafted; see `CHAPTERS.md` for the chapters still to be written.

Software and Tools

- **BLUPF90 Family:** <http://nce.ads.uga.edu/wiki/doku.php>
- **ASReml:** <https://vsni.co.uk/software/asreml-r>
- **R Packages:**
 - **sommer:** <https://cran.r-project.org/package=sommer>
 - **pedigreemm:** <https://cran.r-project.org/package=pedigreeemm>
 - **nadiv:** <https://cran.r-project.org/package=nadiv>
 - **MCMCglmm:** <https://cran.r-project.org/package=MCMCglmm>

Online Resources

- **Animal Breeding and Genetics Wiki:** <http://nce.ads.uga.edu/wiki/>
- **Quarto Documentation:** <https://quarto.org>
- **R for Data Science:** <https://r4ds.had.co.nz/>
- **Mixed Models in R:** <https://m-clark.github.io/mixed-models-with-R/>

Datasets and Examples

All datasets used in this book are synthetic and available in the book repository.

Companion Volumes

Genomic selection and large-scale computation are deferred to separate books; see the Scope section on the welcome page.

Citation Style

This book uses the Genetics citation style. For BibTeX entries, see `references.bib` in the book repository.